



Research Article

Simulation-informed explainable machine learning for evaluating thermochromic and PCM smart glazing in hot climates

Muhammad Huzaifa Zahid^a, Syed Yaqeen Mahdi^a, Muhammad Asif^{a,b,*}^a Department of Architectural Engineering & Construction Management, King Fahd University of Petroleum & Minerals (KFUPM), Dhahran, Saudi Arabia^b Interdisciplinary Research Center for Sustainable Energy Systems (IRC-SES), KFUPM, Dhahran, Saudi Arabia

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ABSTRACT

Solar heat gains through glazing intensify cooling demand in hot climates, yet thermochromic (TC) and phase-change material (PCM) systems are seldom compared under identical conditions within an interpretable predictive framework. This study evaluates TC and nine PCM glazing assemblies in vertical-window and skylight applications under Dhahran conditions and develops an explainable machine-learning surrogate for PCM screening. An EnergyPlus model compared conventional single, double, and triple-clear glazing with the adaptive systems while holding geometry, weather, operation, and HVAC inputs constant. Published monthly cooling profiles validated the conventional-glazing baseline. For each application, 108 PCM-month observations were analyzed using five regression algorithms, a 75/25 train-test split, five-fold cross-validation, and SHapley Additive exPlanations (SHAP). The novelty is a unified workflow that combines validated simulation, application-specific TC-PCM comparison, nonlinear prediction, and feature attribution. Baseline agreement was strong: NMSE ranged from -1.76% to -0.40% and CV(RMSE) from 2.72% to 3.47% for windows, compared with 3.04–3.38% and 4.28–4.65% for skylights. Under the tested conditions, TC glazing reduced annual cooling energy by 41.7–46.7% for windows and 24.8–31.0% for skylights. PCM cooling-energy intensity ranged from 263.49 to 440.65 kWh/m²/year for skylights and from 432.03 to 474.83 kWh/m²/year for windows; Infinite R PCM 25 °C and M51/Q25 yielded the lowest respective values. Gradient Boosting provided the highest test-set accuracy, reaching R² values of 0.970 for skylights and 0.990 for windows. SHAP identified mean outdoor temperature and CDD-24 as the dominant predictors, whereas PCM family and melting temperature showed smaller, application-dependent contributions. The framework enables preliminary screening within the simulated domain and establishes a basis for experimental, hourly, multi-climate, and multi-criteria evaluation.

1. Introduction

Buildings account for a significant share of global energy use, and space cooling is one of the fastest-growing energy end-use sectors,

especially in hot, cooling-dominated climates [1]. Glazing contributes significantly to solar heat gain, increases cooling loads, peak loads, and carbon dioxide emissions [2], and is one of the largest contributors to heat gain in the building envelope [3,4]. While single-, double-, and

List of abbreviations: ACH, Air Changes Per Hour; ANN, Artificial Neural Network; ASHRAE, American Society of Heating, Refrigerating and Air-Conditioning Engineers; CA, Coefficient of Agreement; CDD, Cooling Degree Days; CDD-24, Cooling Degree Days referenced to 24 °C; CFD, Computational Fluid Dynamics; CFL, Compact Fluorescent Lighting; CNN, Convolutional Neural Network; COP, Coefficient of Performance; CSP, Concentrating Solar Power; CV, Cross-Validation; CVRMSE, Coefficient of Variation of the Root Mean Square Error; DL, Deep Learning; EPW, EnergyPlus Weather; ET, Extra Trees Extremely Randomized Trees; EUI, Energy Use Intensity; GA, Genetic Algorithm; GB, Gradient Boosting; HVAC, Heating, Ventilation, and Air Conditioning; IRC-SES, Interdisciplinary Research Center for Sustainable Energy Systems; K-fold CV, k-Fold Cross-Validation; K-NN, k-Nearest Neighbors; KFUPM, King Fahd University of Petroleum & Minerals; KGE, Kling-Gupta Efficiency; kWh, Kilowatt-hour; LR, Linear Regression; LSTM, Long Short-Term Memory; MAE, Mean Absolute Error; ML, Machine Learning; MLP, Multi-Layer Perceptron; NMSE, Normalized Mean Bias Error; NSE, Nash-Sutcliffe Efficiency; PBIAS, Percent Bias; PCM, Phase Change Material; PV, Photovoltaic; R², Coefficient of Determination; RF, Random Forest; RH, Relative Humidity; RH%, Relative Humidity percentage; RL, Reinforcement Learning; RMSE, Root Mean Square Error; SHAP, Shapley Additive explanations; SVM, Support Vector Machine; SVR, Support Vector Regression; TC, Thermochromic coating; T_{lum}, Luminous Transmittance; VO₂, Vanadium Dioxide; WWR, Window-to-Wall Ratio; XAI, Explainable Artificial Intelligence; XG-Boost, Extreme Gradient Boosting; ΔT_{sol}, Solar Modulation Efficiency

* Corresponding author at: Department of Architectural Engineering & Construction Management, King Fahd University of Petroleum & Minerals (KFUPM), Dhahran, Saudi Arabia.

E-mail address: asifm@kfupm.edu.sa (M. Asif).

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triple-glazing systems are good for thermal insulation, their optical and thermal characteristics are largely fixed and cannot adapt to the ever-changing environment [5]. As a result, a number of smart glazing solutions are emerging to dynamically control solar gain while maintaining building energy efficiency and indoor thermal comfort [3].

Thermochromic (TC) glazing and phase change material (PCM) glazing are two emerging technologies for smart envelopes that have gained significant attention for their dynamic ability to control solar heat gain and thermal energy transfer [3,6,7]. Thermochromic glazing changes its optical characteristics with temperature and reduces solar radiation transmission when thermal stress is high but provides daylight when thermal stress is low [7]. PCM-integrated glazing, on the other hand, can store and release latent heat during phase changes, helping regulate indoor temperature and reduce peak cooling demand [8,9]. Given these complementary mechanisms, both technologies have shown great potential to improve the energy performance of buildings, reduce energy use for building cooling, and enhance indoor thermal comfort. In this sense, they have proven promising solutions for designing climate-responsive, energy-efficient building envelopes, especially in hot climates where cooling demand accounts for a major share of a building's energy use year-round [3,8,10].

Although significant advances have been made in research on thermochromic and PCM glazing, several challenges remain [7]. To date, most of these technologies have been studied separately or with different system models; it is difficult to make direct comparisons of their performance, especially in hot and cooling-dominated climates [11,12]. Furthermore, past studies have largely relied on building energy simulations, which are time-consuming and do not enable rapid testing of various glazing combinations in the early design phase [13,14]. While machine-learning methods have been applied in recent years to develop building energy applications, their use in conjunction with validated building energy simulations for assessing smart glazing has not yet been widely adopted, and the relative contributions of climatic and material parameters to cooling performance remain poorly understood [15,16]. The identified research gaps underscore the need for a comprehensive framework that integrates validated building energy simulation modeling with explainable machine learning methods to enable accurate, explainable, and efficient assessment of state-of-the-art smart glazing technologies.

To address these research gaps, a thorough simulation-based, explainable machine-learning model is developed to assess the energy performance of thermochromic (TC) and phase-change material (PCM) smart glazing systems in hot climates. A model rigorously validated using EnergyPlus software compares the performance of conventional single, double, and triple-clear glazing with thermochromic glazing and nine PCM-integrated glazing configurations for both skylight and vertical window applications. In addition to building energy simulation, machine-learning-based surrogate models are trained to quickly predict cooling performance, and SHAP is used to quantify the contributions of climatic parameters and material characteristics to cooling demand. The suggested framework combines high-fidelity simulation and explainable artificial intelligence to provide a reliable, interpretable, and computationally efficient method for assessing and selecting climate-responsive smart glazing technologies.

This study develops and evaluates an explainable, simulation-informed machine-learning framework for selecting thermochromic and PCM-integrated glazing in a hot, cooling-dominated climate. A validated EnergyPlus model quantifies cooling loads for skylights and vertical windows under consistent conditions. The framework compares conventional, thermochromic, and nine PCM-integrated glazing configurations; evaluates five regression algorithms; and combines statistical testing, Taylor diagrams, and SHAP analysis to assess prediction reliability and identify governing variables. The results provide application-specific guidance for reducing cooling energy. The current study pursues three distinct objectives:

- Validate the EnergyPlus model against published benchmarks and ASHRAE Guideline 14 criteria, then compare the cooling performance of conventional, thermochromic, and PCM-integrated glazing for skylights and vertical windows.

- Develop and evaluate Linear Regression, Random Forest, Gradient Boosting, Extra Trees, and k-Nearest Neighbors using test-set metrics, five-fold cross-validation, Kruskal-Wallis testing, and normalized Taylor diagrams.
- Use SHAP analysis to identify the climatic and material variables governing predictions and translate the combined simulation and machine-learning results into application-specific glazing-selection guidance.

Existing studies rarely compare thermochromic and PCM glazing in a single validated model under identical conditions for both skylights and vertical windows, and seldom resolve the coupled effects of climate and PCM properties on cooling demand. This study addresses these gaps using validated EnergyPlus simulations of conventional, thermochromic, and nine PCM glazing configurations under Dhahran conditions, integrated with five machine-learning models, seven performance metrics, five-fold cross-validation, Kruskal-Wallis testing, Taylor analysis, and SHAP attribution. The framework quantifies application-specific cooling performance, identifies dominant climatic and material controls, and provides an interpretable surrogate for rapid, simulation-consistent façade and PCM selection in cooling-dominated buildings.

2. Literature review

The recent development of smart glazing technologies has proven to be a highly promising solution for controlling solar heat gain and daylight penetration in buildings, with the objective of enhancing energy efficiency [6,7,10]. The research has mainly focused on thermochromic (TC) glazing, phase change material (PCM)-integrated glazing, adaptive façade systems, and simulation-based performance evaluation across different climates [17]. Recently, optimization techniques and machine learning have become effective tools for rapid design and prediction of glazing performance [18,19]. The use of explainable artificial intelligence (XAI), the comparison of the performance of different smart glazing technologies under identical conditions, and the simultaneous assessment of window and skylight applications are still not widely adopted [20,21]. In order to give a quantitative impression of the main features of the research that have been identified in the selected representative studies, the distribution of these features is summarized in Fig. 1. The bubble map shows the relative importance of smart glazing technologies, methodological approaches, and application domains, with each bubble's size corresponding to the number of studies that discuss the technology, methodology, or domain. This visualization enables a quick analysis of existing research priorities and serves as a basis for identifying underexplored areas that inspire current research. Fig. 2 maps keyword co-occurrence and average publication year across the analyzed records from 2016 to 2024. Node size reflects keyword occurrence. Spatial proximity and link thickness indicate co-occurrence strength. The color scale progresses from purple for earlier records to yellow for recent records. The keywords “review,” “building energy performance,” and “smart windows” occupy the network core. Machine learning, building energy consumption, and thermal performance form major connected themes. Earlier records emphasize thin films, cooling load, and adaptive glazing. Recent records shift toward machine learning, SHAP, window properties, energy policy, ionic polymer, and thermochromic hydrogel. This progression aligns with the present study's integration of smart-glazing simulation, machine-learning prediction, and SHAP-based interpretation.

Table 1 presents a comparative synthesis of representative studies across glazing technology, simulation methodology, intelligent modeling, application type, research focus, and research limitations to systematically identify the current state of research, technological progress, and existing research gaps. This evidence-based comparison underpins further quantitative trend analysis and identifies unresolved research challenges that the present study addresses.

Table 2 links building-oriented PCM and thermochromic modeling with broader evidence from photovoltaic, battery, solar-energy, and heat-storage applications. The machine-learning studies demonstrate

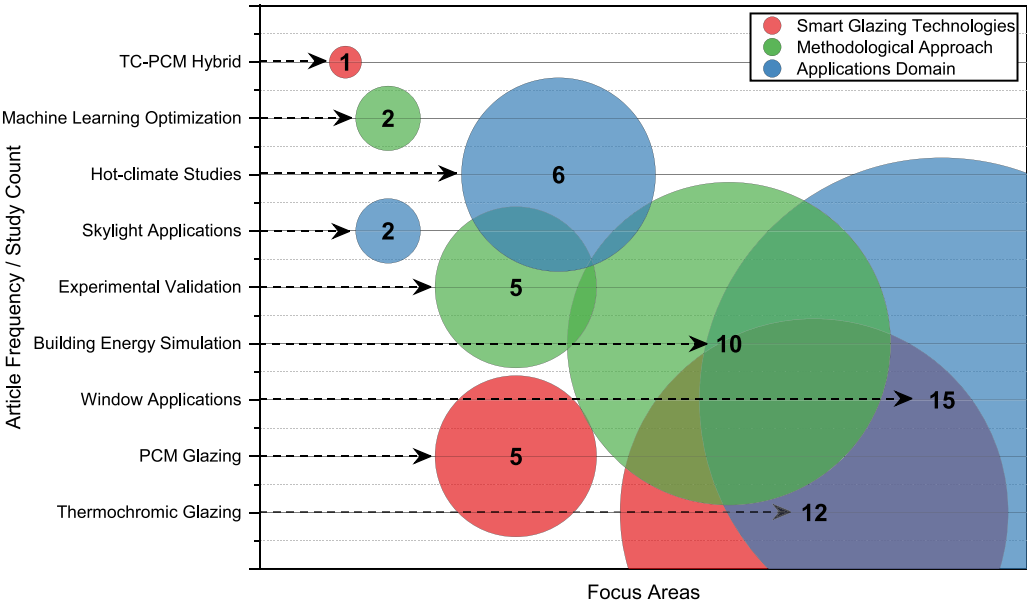


Fig. 1. Distribution of key research characteristics identified from the selected representative studies.

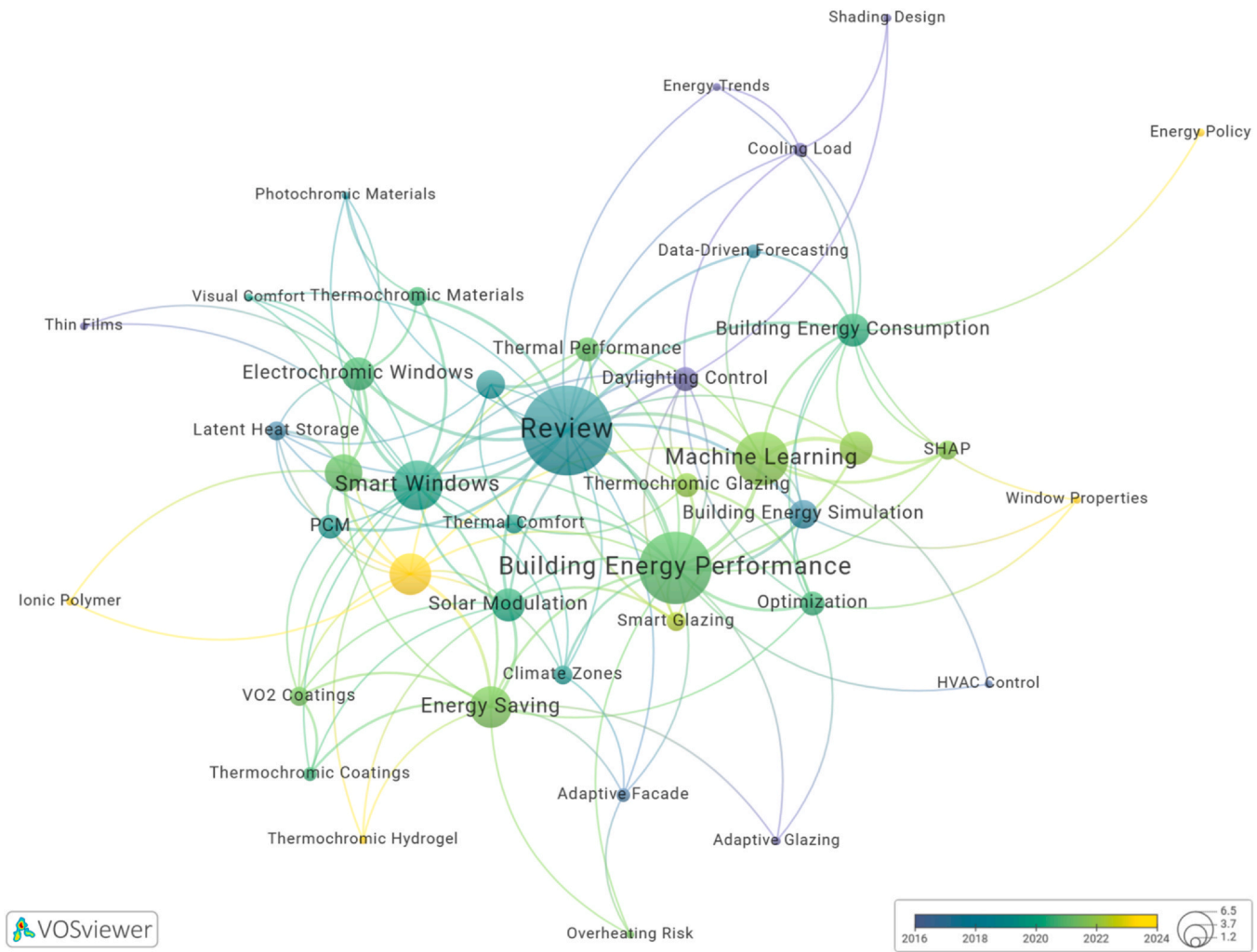


Fig. 2. VOSviewer map of keyword co-occurrence and average publication year from 2016 to 2024.

Table 1
Comparative synthesis of representative studies.

ID	Title	TC	PCM	Energy Simulation	Machine Learning	Explainable AI	Window	Skylight	Hot Climate	Research Focus	Research Limitations	Study Ref.
1	Solar Modulation Utilizing VO2-Based Thermochromic Coatings for Energy-Saving Applications.	✓	✗	Review	✗	✗	✓	✗	Mixed	Material Development	No Simulation; No PCM; No ML; No XAI	Cao et al. [22]
2	Multi-objective Optimization of Thermochromic Glazing Properties to Enhance Building Energy Performance.	✓	✗	EnergyPlus	Surrogate	✗	✓	✗	✓	Optimization	No PCM; No XAI; No Skylight	Araújo et al. [23]
3	Numerical Analysis on the Thermal Performance of PCM-Integrated Thermochromic Glazing Systems.	✓	✓	Numerical	✗	✗	✓	✗	Mixed	Hybrid Glazing	No ML; No XAI; No Skylight	Jin et al. [24]
4	Energy Performance and Comfort Analysis of Three Thermochromic Glazing Materials.	✓	✗	EnergyPlus	✗	✗	✓	✓	✓	Thermal Comfort	No PCM; No ML; No XAI	Tennakoon et al. [25]
5	Energy Performance of Thermochromic Smart Windows in Different Climate Zones.	✓	✗	EnergyPlus	✗	✗	✓	✗	✓	Climate Assessment	No PCM; No ML; No XAI	Wu et al. [26]
6	Comparison of Thermal Energy Saving Potential of Four Adaptive Façade Technologies.	✓	✓	DesignBuilder / EnergyPlus	✗	✗	✓	✗	✗	Façade Comparison	No ML; No XAI; No Skylight	Attia et al. [27]
7	Realistic Glazing/Window Properties for Minimizing Building Energy Use Through Simulation-Based Optimization.	✓	✗	EnergyPlus	Optimization	✗	✓	✗	Multiple	Optimization	No PCM; No XAI; No Skylight	Lu et al. [28]
8	Thermal Performance of a PCM-Filled Double-Glazing Unit with Different Thermophysical Parameters of PCM.	✗	✓	Numerical	✗	✗	✓	✗	✗	PCM Performance	No TC; No ML; No XAI	Li et al. [29]
9	Prospects and Challenges of PCMs Integrated into Building Glazing Systems.	✗	✓	Review	✗	✗	✓	✗	Mixed	Review	No TC; No ML; No XAI	Chi et al. [30]
10	Energy and Thermal Performance Analysis of PCM-Incorporated Glazing Units Combined with Passive and Active Techniques.	✗	✓	Review	✗	✗	✓	✓	Mixed	Review	No TC; No ML; No XAI	Arasteh et al. [12]
11	Design and Scalable Synthesis of Thermochromic VO2-Based Coatings for Energy-Saving Smart Windows.	✓	✗	Experimental	✗	✗	✓	✗	✗	Material Development	No Building Simulation; No ML; No XAI	Kaufman et al. [31]
12	A highly visible-transparent thermochromic smart window with broadband infrared modulation for all-season energy savings.	✓	✗	Building Simulation	✗	✗	✓	✗	Multiple	Smart Window Design	No PCM; No ML; No XAI	Jiang et al. [32]
13	Hydrated Ionic Polymer for Thermochromic Smart Windows in Buildings.	✓	✗	Energy Simulation + Field Test	✗	✗	✓	✗	✓	Experimental Validation	No PCM; No ML; No XAI	Wang et al. [33]
14	Three-State Thermochromic Smart Window for Building Energy-Saving.	✓	✗	Building Simulation	✗	✗	✓	✗	Multiple	Smart Window Design	No PCM; No XAI; No ML	Liu et al. [34]
15	Energy-Efficient Smart Window Based on a Thermochromic Hydrogel with Adjustable Critical Response Temperature.	✓	✗	Indoor Simulation	✗	✗	✓	✗	✓	Hydrogel Smart Window	No PCM; No ML; No XAI	Sun et al. [35]

Table 2
Representative recent studies combining PCM or TC glazing/coating technologies with machine learning or deep learning methods.

ID	Title	Technology	ML / DL Approach	Application Focus	Key Finding	Reference
1	Multi-objective optimization of cooling and heating loads in residential buildings integrated with phase change materials using the artificial neural network and genetic algorithm.	PCM	ANN + Genetic Algorithm	Multi-objective optimization of PCM melting point/thickness for residential cooling & heating loads	Optimal melting point/thickness trade-off identified (ANN-GA).	Bagheri-Esfah et al. [36]
2	A machine learning and deep learning-based approach to predict the thermal performance of phase change material integrated building envelope.	PCM	ML & Deep Learning (ANN, SVM, DL)	Predicting thermal performance of PCM-integrated building envelope	DL models outperform ANN/SVM benchmarks.	Bhamare et al. [37]
3	Forecasting energy demand of PCM-integrated residential buildings: A machine learning approach	PCM	ML (regression/ensemble)	Forecasting energy demand of PCM-integrated residential buildings	Accurate multi-climate energy-demand forecasts.	Zhussupbekov et al. [38]
4	Machine learning-based predicting of PCM-integrated building thermal performance: An application under various weather conditions in Morocco	PCM	Machine Learning	Predicting PCM-integrated building thermal performance across Moroccan climate zones	Strong generalization across 3 climate zones.	Jraida et al. [39]
5	Assessing the performance of a Trombe wall enhanced with phase change material using deep learning	PCM	Deep Learning (CNN + LSTM)	Predicting thermal efficiency/heat flux of a PCM-enhanced Trombe wall	CNN-LSTM: $R^2 = 0.999$ (best of 8 models).	Krechowicz et al. [40]
6	Unique dielectric dispersion induced ultra-broadband microwave absorption of tellurium-doped black phosphorus nanoflakes/aramid nanofibers/carbonyl iron nanopowders ultra-lightweight aerogel.	TC (VO ₂)	ML-guided synthesis (regression)	Optimizing hydrothermal synthesis parameters for thermochromic VO ₂ nanoparticles	ML-guided synthesis to pure-phase VO ₂ .	Wang et al. [41]
7	Preparation of Thermochromic Vanadium Dioxide Films Assisted by Machine Learning	TC (VO ₂)	ML (MLP, RF, SVM, XG-Boost)	Optimizing magnetron-sputtering deposition parameters for thermochromic VO ₂ films	XG-Boost: 88.5% accuracy (pure-phase VO ₂ (M)).	Xiong et al. [42]
8	Optimizing transition temperatures for thermochromic smart windows via neural networks coupling simulation	TC	Neural Network + simulation	Optimizing transition temperature of thermochromic smart windows	Climate-optimal switching temperature identified.	Lu et al. [43]
9	Global and regional perspectives on optimizing thermo-responsive dynamic windows for energy-efficient buildings	TC	ANN (multi-layer perceptron)	Global/regional energy-performance mapping of thermo-responsive dynamic windows	ANN on 2.8 M simulations; open-source tool (Py-Dynamic-Window).	Gao et al. [44]
10	Mechanical vibration's effect on thermal management module of Li-ion battery module based on phase change material (PCM) in a high-temperature environment	N-octadecane PCM	no ML	Vibration effects on battery temperature at 323 K	At 1 C, 2 C, and 3 C, vibration reduced ΔT to 0.66, 2.28, and 4.61 K, representing reductions of 57.4%, 53.8%, and 58.0%. Optimum: 5-mm PCM, 40-mm amplitude and approximately 80 Hz.	Shen et al. [45]
11	Efficient approaches for building-integrated photovoltaic modules:Advancements in phase change materials and fin designs	RT28HC PCM with copper fins; CFD	no ML	PV cooling, PCM melting and electrical efficiency	Eight Y-fins achieved a 0.93 liquid fraction, 53.98 W heat-storage rate and 16.77% electrical efficiency after 3600 s. The reported efficiency improvement was 3.1%.	Kaplan et al. [46]
12	Phase change materials in thermal management of Li-ion batteries: Astate-of-the-art review	Pure, nano-enhanced and hybrid PCMs	Review of ANN, RF, SVR, GA and RL	Battery cooling and intelligent optimization	Reviewed hybrids reduced temperature by approximately 10–20 °C for PCM-air and over 20 °C for PCM-liquid/heat-pipe systems. A reviewed ANN-GA design reduced maximum temperature by 15%.	Talebizadehsardari et al. [47]
13	Numerical investigation of the impact of toothed fins on the heat transferperformance of a shell-and-tube exchanger during phase change materialmelting process	Paraffin PCM with annular and longitudinal toothed fins; 3D CFD	no ML	Fin-volume and tooth-spacing optimization	A 20% reduction in fin volume caused only approximately 1% loss in melting enhancement. Recommended opening: 10–20%, with 16 annular teeth or 1-mm longitudinal spacing.	Wu et al. [48]
14	Harnessing solar energy with phase change materials: A review of meltingpoint impacts	Organic, inorganic and eutectic PCMs; narrative review	no ML	PCM selection for solar thermal, PV and CSP systems	Reviewed RT35HC application reduced PV temperature by 24.9 °C and increased power by 11.02%. Typical conductivity ranges were 0.1–0.5 W/m.K for organic and 0.5–2.0 W/m.K for inorganic PCMs.	Togun et al. [49]

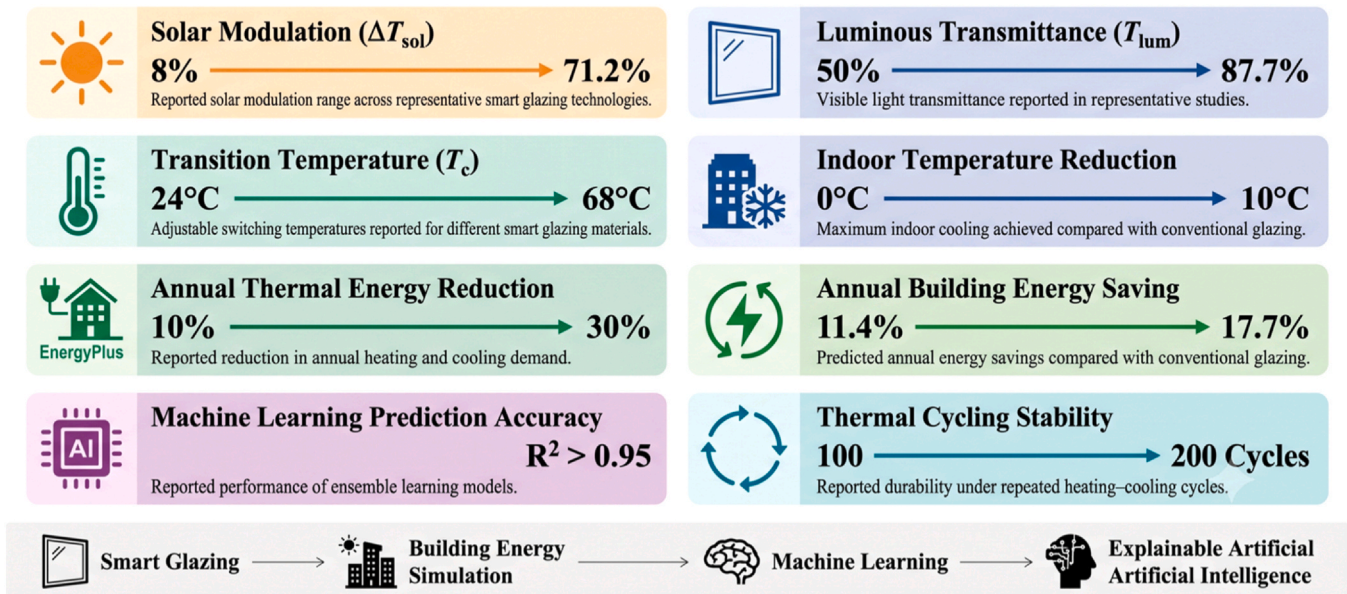


Fig. 3. Performance metrics and energy reduction ranges of smart glazing systems across existing studies.

the use of neural networks, ensemble methods, and deep learning for predicting envelope performance, optimizing PCM parameters, and controlling thermochromic transition behavior. The mechanism-focused studies identify melting-point compatibility, latent-heat storage, low thermal conductivity, heat-transfer enhancement, and operating conditions as major determinants of PCM charging and thermal regulation. These studies strengthen the thermophysical basis for examining PCM configuration and melting temperature. The comparison also confirms the unresolved research gap: previous studies rarely integrate thermochromic and PCM assessment for both vertical windows and skylights with whole-building simulation, multiple regression models, and SHAP-based interpretation under a hot climate.

2.1. Quantitative assessment of smart glazing performance

Existing literature indicates that smart glazing technologies have demonstrated reliable results in enhancing building energy performance through dynamic control of solar heat gain without compromising the building's thermal environment [7,10]. Thermochromic glazing remains a major research focus due to its passive nature and ability to integrate into the building envelope [50,51]. The values of solar modulation ΔT_{sol} range from about 8% to above 70% for different materials and engineering approaches, while the values of luminous transmittance T_{lum} range from 50% to 90% depending on material composition and engineering strategy, and the engineering and material composition can be adjusted to achieve different transition temperatures, ranging from 24 °C to 68 °C [35,52]. The advanced hydrogel-based thermochromic windows have achieved ΔT_{sol} as high as 71.2% and T_{lum} as high as 87.5%, with a lower critical solution temperature of 24.1 °C and an upper critical solution temperature of 33.2 °C, and exhibit excellent cyclic durability of more than 100 thermal cycles [35]. Likewise, the ionically conductive windows with polymer in water demonstrated the following properties: $\Delta T_{sol} = 30.5\%$, $T_{lum} = 87.7\%$, temperature range of changes = 25–42 °C, no degradation after 200 heating-cooling cycles, and a temperature drop of 10 °C in the interior in summer. Simulation results indicated an annual energy saving of 11.4–17.7% over the conventional glazing [32,53].

Fig. 3 summarizes the key performance indicators reviewed in smart glazing studies, including solar modulation (ΔT_{sol}), luminous transmittance (T_{lum}), transition temperature, indoor temperature reduction, annual energy savings, machine learning accuracy, and durability. The figure shows the range of performance achieved in recent periods,

driven by innovations in materials and sophisticated evaluation techniques, providing a comprehensive picture of the current state of smart glazing technologies. To assess the energy performance of smart glazing across various climates, building energy simulation is the primary tool used [7,54,55]. In most simulation studies, annual cooling, heating, and thermal comfort performance are quantified using EnergyPlus or similar whole-building simulation platforms [17]. Reported annual thermal energy demand reductions typically ranged from 10% to 30% (with better results for advanced adaptive façade technologies and their ability to mitigate overheating, depending on the control strategy and climate) [7,56]. Comparative studies also revealed that dynamic shading and electrochromic glazing provided the highest annual energy saving in temperate office buildings, while in colder climates, adaptive ventilated façades and double-skin systems are more beneficial [7,17,57]. Along with the trend towards multifunctional solutions, the design of such materials has shifted towards VO₂ doping, hydrogel composites, core-shell nanostructures, multilayer optical coatings, and thermochromic systems that combine the advantages of luminous transmittance, near-infrared modulation, durability, and stability [58–60].

More recently, machine learning has been used as a complementary approach to speed up material optimization and performance prediction [18,19,61]. Some ensemble-based algorithms, such as Random Forest, Gradient Boosting, Extra Trees, and XG-Boost, have been shown to be highly predictive, with reported R-squared values often above 0.95 for predicting material properties [15]. Efforts to increase transparency in the models have greatly benefited from explainable AI (XAI) approaches such as SHAP, feature importance analysis, and partial dependence plots [62,63]. The use of explainable AI (XAI) techniques, including SHAP, feature importance analysis, and partial dependence plots, has been a pivotal step in enhancing model transparency and pinpointing the most significant variables driving performance [21,63]. However, the use of interpretable machine learning remains limited to materials engineering and concrete optimization, while physics-based simulation and explainable, data-driven prediction for smart building envelope design have significant potential [15,64].

2.2. Research gaps and future opportunities

Although a significant advance has been made in the study of smart glazing, there remain some major limitations:

- Thermochromic glazing predominated in the studies: Most studies examined thermochromic glazing; however, fewer examined PCM glazing or hybrid thermochromic-PCM systems under similar conditions [7,50].
- Limited comparative investigations: Most existing studies have evaluated a single type of glazing technology, making it difficult to directly compare their performance [7,65].
- Window-dominated applications: Most investigations have centered on vertical window systems, and skylight applications remain under-researched despite their higher solar gain and cooling contribution [5,8].
- Limited validation methodologies: Most existing literature omitted comprehensive validation, such as integrated simulations, experimental validation, or long-term field testing, as only a few studies deployed these rigorous methods [12,66].
- Climatic evaluations are limited: There is limited generalizability because most investigations examined one or a small number of climatic conditions.
- Most existing studies overlooked annual building energy performance and thermal comfort, focusing instead on localized material characterization. Hence, prior research prioritized isolated properties like luminous transmittance T_{lum} , solar modulation efficiency ΔT_{sol} , phase transition temperatures, and basic optical traits rather than full building-scale assessments. [7,50,60].
- Low machine learning adoption: Machine learning has proven very effective at predicting material performance but is not widely used to assess smart glazing performance [19].
- Lack of Explainable Artificial Intelligence (XAI): None of the studies on glazing employed any explainable machine learning methods, such as SHAP or feature attribution, to enhance model transparency and support engineering decision-making [62,63].
- Lack of optimization and decision-support frameworks: Limited research consisted of building energy simulations and interpretable predictive models to enable fast decision-making and optimization of smart glazing technologies [19,56].
- Economic feasibility, life-cycle cost analysis, and practical implementation considerations are rarely added to the energy-performance assessment [7,50].

The evidence analyzed shows how smart glazing technologies have developed rapidly and highlights the need for integrated, transparent, and data-driven evaluation frameworks.

3. Methodology

The present study employs a multi-phase, structured methodology to evaluate the performance of the integrated TC-PCM glazing system in a hot climate. The workflow consists of literature review, model development, simulation setup, baseline validation, and performance assessment (as illustrated in Fig. 4). This systematic approach ensures analytical quality and reliability in evaluating the adaptive glazing system.

3.1. Baseline model setup

The EnergyPlus case model represents a 10 m × 10 m square building with 100 m² floor area, 3.5 m ceiling height, 85% window-to-wall ratio, and an orientation of 245° from geographic north. Dhahran EPW data drive the simulation. Boundary conditions include 0.3 ACH infiltration, 3 ACH mechanical ventilation, 0.0215 persons/m² occupancy, 117 W/person internal heat gain, 5 W/m² lighting, and continuously operating split-unit HVAC. EnergyPlus calculates cooling demand from envelope conduction, solar transmission, infiltration, ventilation, internal gains, and HVAC response. Single, double, and triple-clear glazing establish the benchmarks. Subsequent runs replace only the glazing assembly with thermochromic and nine PCM systems,

while geometry, weather, schedules, loads, and HVAC settings remain fixed. Table 3 reports on the complete boundary conditions.

EnergyPlus, the industry leader in simulation, developed the initial building model based on traditional glazing systems. This setting serves as the system reference when no other settings are specified.

3.2. Simulation model validation

Baseline validation used the monthly cooling-electricity profiles reported by Mohammed et al. [67] for conventional 3 mm single-clear, 3 mm double-clear with a 6 mm air gap, and 3 mm triple-clear glazing with 6 mm air gaps. The reference study reported annual cooling energy of 39,565.18 kWh for wall-mounted 3 mm single-clear glazing and 63,340.74 kWh for the corresponding skylight. The present model reproduced the published geometry, climate, envelope, occupancy, infiltration, HVAC, and glazing conditions. ASHRAE Guideline 14 [68] criteria governed the model accuracy evaluation, utilizing Normalized Mean Bias Error (NMBE, Eq.1) and the Coefficient of Variation of the RMSE [CV(RMSE), Eq.2]. The validation process required maintaining the NMBE within ± 5% and the CV(RMSE) below ± 15%, establishing a strong benchmark for evaluating thermochromic and PCM smart glazing systems in further studies.

Where:

- m_i represents the actual value of the measurement (or observed data) for the i^{th} observation,
- s_i denotes the corresponding predicted (or simulated) value.
- n refers to the total number of data points or observations considered in the analysis.
- m is the mean of all actual values, calculated by summing all m values and dividing by the total number of observations, i.e., $(m = \sum m_i / n)$.
- p indicates the number of parameters used in the prediction model, typically 1 in straightforward cases where only one variable is predicted.

$$NMBE = \frac{1}{m} \cdot \sum_{i=1}^n \left(\frac{m_i - s_i}{m_i} \right) \times 100(\%) \quad (1)$$

$$CV(RMSE) = \frac{RMSE}{\bar{m}} = \frac{\sqrt{\frac{\sum (m_i - s_i)^2}{n}}}{\frac{\sum m_i}{n}} \times 100 \quad (2)$$

3.3. Thermochromic and PCM glazing configurations and parametrization

The investigated glazing systems are multilayer glass systems with a thermochromic (TC) coating or a PCM layer incorporated between the outer and inner glass panes. For the thermochromic glazing, the active TC coating enabled temperature-dependent solar modulation, yielding a total thermal transmittance (U-value) of 2.130 W/m².K. Each PCM assembly contained an 11.2 mm PCM layer between two 3 mm polycarbonate sheets, giving a total thickness of 17.2 mm. The nine configurations comprised six BioPCM grades, two Infinite R products, and one WINCO ENERCEL product. Table 4 reports nominal thermal conductance without surface-film resistances rather than whole-assembly U-value.

The EnergyPlus definitions assigned each PCM product its thermal conductivity, density, specific heat, and temperature-enthalpy relationship, with latent heat and the phase-change interval represented in the enthalpy curve; the Finite Difference algorithm then simulated transient melting, solidification, energy storage, and heat release. The machine-learning models used melting temperature as a numerical feature and material family as a one-hot-encoded categorical feature, whereas conductivity, latent heat, density, specific heat, and conductance were not included independently, although their combined effects influenced the EnergyPlus cooling-load targets. Therefore,

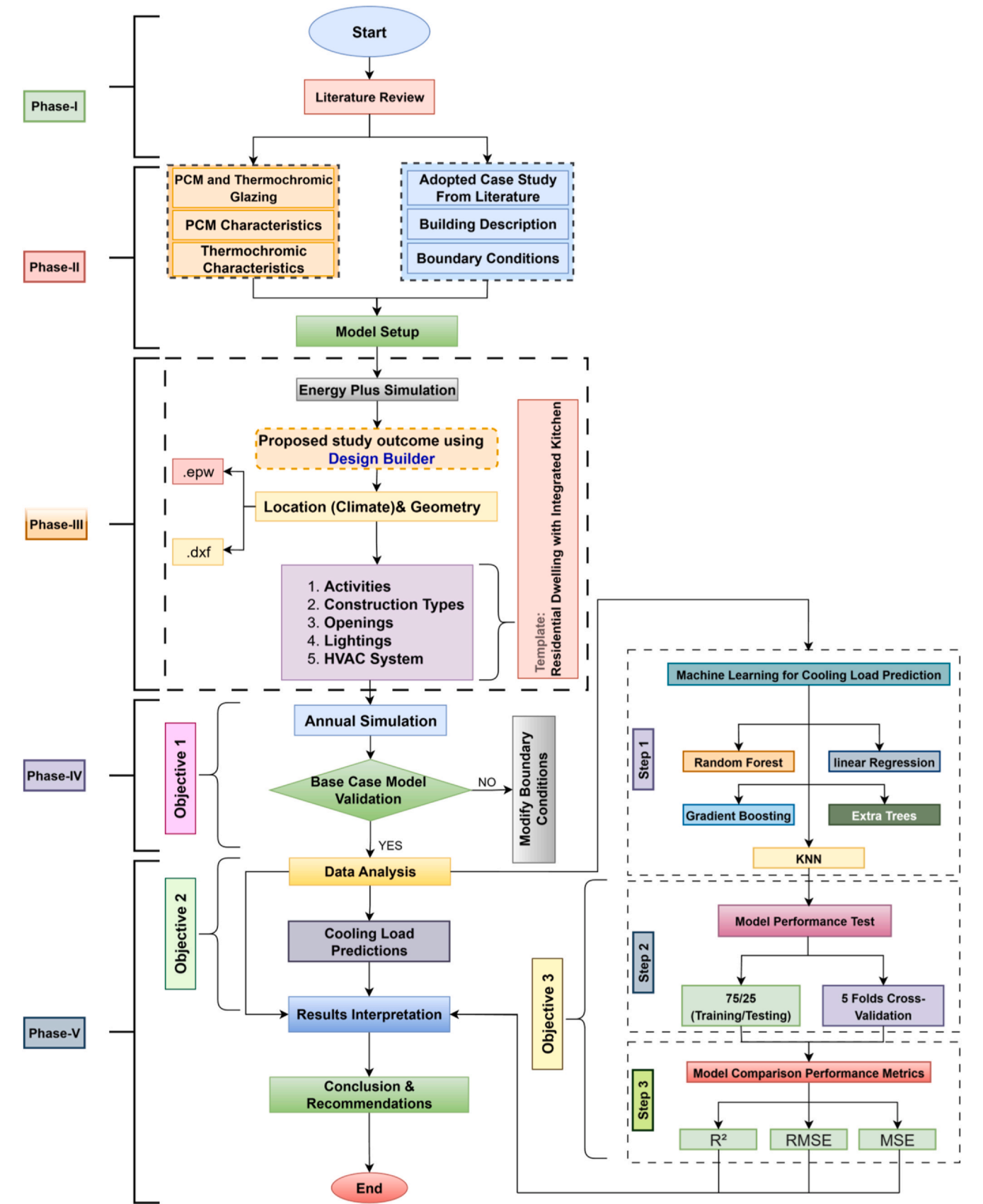


Fig. 4. Methodology flowchart.

Table 3
Boundary conditions used in the EnergyPlus simulation.

Parameter	Model Description
Location	Dhahran, Saudi Arabia (26.27° N, 50.15° E)
Building Orientation	Positioned at 245° from Geographical North
Layout Shape	Square configuration
Ceiling Height	3.5 m
Total Floor Area	100 square meters (10 m × 10 m)
Window-to-Wall Ratio	85%
Infiltration Rate	0.3 air changes per hour (ACH)
Occupant Density	0.0215 persons per square meter
Internal Heat Gains	117 W per person
Clothing Insulation Levels	0.5 clo (summer); 1 clo (winter)
Wall Thermal Transmittance	0.351 W/m ² ·K
Roof Thermal Transmittance	0.346 W/m ² ·K
Window U-Values	Case-specific (see individual window specification above)
HVAC system configuration	Split unit with independent mechanical ventilation
Ventilation Rate	3 ACH via mechanical system
Supply Air Temperature	12 °C
System Operation Mode	Operates continuously
Cooling Setpoint (Summer)	24 °C
Lighting Power Density	5 W/m ² with target illumination of 100 lx
Lighting Type	Compact fluorescent lighting (CFL)
Heating Setpoints	Active heating: 18 °C; Setback: 12 °C
Cooling Setpoints	Active cooling: 21 °C; Setback: 28 °C

configurations sharing the same family and melting temperature received identical machine-learning descriptors, limiting the models to interpolation within the evaluated configurations and preventing independent attribution of performance to individual thermophysical properties. The results should therefore not be interpreted as evidence of transferable property-performance relationships or reliable extrapolation. Such interpretation requires a larger factorial dataset that varies conductivity, latent heat, density, specific heat, and thickness independently, together with configuration-level or grouped validation using unseen PCM combinations. Fig. 5 illustrates the detailed glazing and skylight assembly workflow. The simulations ran at a 100% W/W ratio with maximum solar access, which clearly isolates the thermal performance of the advanced glazing systems.

3.4. Machine learning-based performance analysis

The workflow aggregated monthly cooling-load outputs from validated EnergyPlus simulations for nine PCM configurations in skylight and vertical-window applications. Each application dataset contained 108 observations, comprising nine PCM systems across 12 months. Cooling load served as the target variable. Climate variables, sine and cosine month encodings, PCM melting temperature, and PCM family served as predictors. The method applied a randomized 75/25 train-test

split with “random_state=42”, assigning 81 observations to training and 27 to testing. A shuffled five-fold cross-validation procedure with “random_state=42” provided an additional evaluation. The analysis assessed Linear Regression, Random Forest, Gradient Boosting, Extra Trees, and k-Nearest Neighbors using R², RMSE, and MAE.

Separate regression pipelines evaluated skylight and vertical window cooling loads. Each pipeline predicts monthly cooling energy (kWh) from seven numerical variables: mean outdoor temperature, relative humidity, sun hours, CDD-24, sine and cosine month terms, and PCM melting temperature, together with the categorical PCM family variable. The pipeline retained numerical variables on their original scales, whereas a one-hot encoder mapped the PCM family feature with a designated reference category. The workflow applied identical pre-processing and model settings to both applications, while the training routine fitted the preprocessing stage strictly within each partition to prevent information leakage.

The five models represent complementary response structures. Linear Regression estimates an additive linear relationship. Random Forest averages predictions from bootstrap trees. Extra Trees averages predictions from trees generated with randomized split thresholds. Gradient Boosting sequentially fits depth 3 trees to residuals. k-Nearest Neighbors predicts from the distance-weighted mean of the five closest training observations. The models were evaluated using the held-out test set and shuffled five-fold cross-validation with “random_state=42”. Dataset size, training and testing allocation, and model-specific hyperparameters.

The Kruskal-Wallis H test compared six test-set groups: actual cooling loads and predictions from Linear Regression, Random Forest, Gradient Boosting, Extra Trees, and k-Nearest Neighbors. Each group contained 27 observations. Alpha is set to ($\alpha = 0.05$). The null hypothesis specified no difference among the six rank distributions. The H test evaluates ranked distributions rather than exact equality of arithmetic means. Therefore, the reported arithmetic mean and relative mean bias are alongside H and p. The methodological design pooled a six-ranked group featuring ($k = 6$), ($n_j = 27$), ($N = 162$), and ($k - 1 = 5$) degrees of freedom. In Eq. (3), R_j denotes the rank sum for group j , n_j denotes its sample size, and t_l denotes the number of observations within tied-rank set l . The denominator applies the tie correction. A p-value above $\alpha = 0.05$ indicates no statistically detectable rank-distribution difference. The test does not establish equality of arithmetic means, so group means and relative means bias are quantified separately [69].

$$H_c = \frac{\frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} - 3(N+1)}{1 - \frac{\sum_{l=1}^m (t_l^3 - t_l)}{N^3 - N}} \quad (3)$$

To compare the predictive behavior of the five algorithms, normalized Taylor diagrams are constructed from the 27 test observations for each application. Three performance metrics are assessed for each model: the Pearson correlation coefficient between actual and predicted cooling loads, the ratio of predicted to observed standard deviations,

Table 4
Thermal transmittance (U-value) of thermochromic and PCM-integrated glazing and skylight systems.

PCM assembly	Existing no-film conductance (W/m ² ·K)	Layer resistance (m ² ·K/W)	Corrected EN 673 U-value (W/m ² ·K)
PCM M27/Q27	11.82	0.08460	3.930
PCM M27/Q29	11.35	0.08811	3.876
PCM M51/Q25	10.94	0.09141	3.827
PCM M51/Q27	10.52	0.09506	3.775
PCM M91/Q25	10.88	0.09191	3.820
PCM M91/Q27	10.47	0.09551	3.768
Infinite R PCM 23 °C	9.96	0.10040	3.700
Infinite R PCM 25 °C	9.52	0.10504	3.638
WINCO ENERCEL 23 °C	9.88	0.10121	3.689

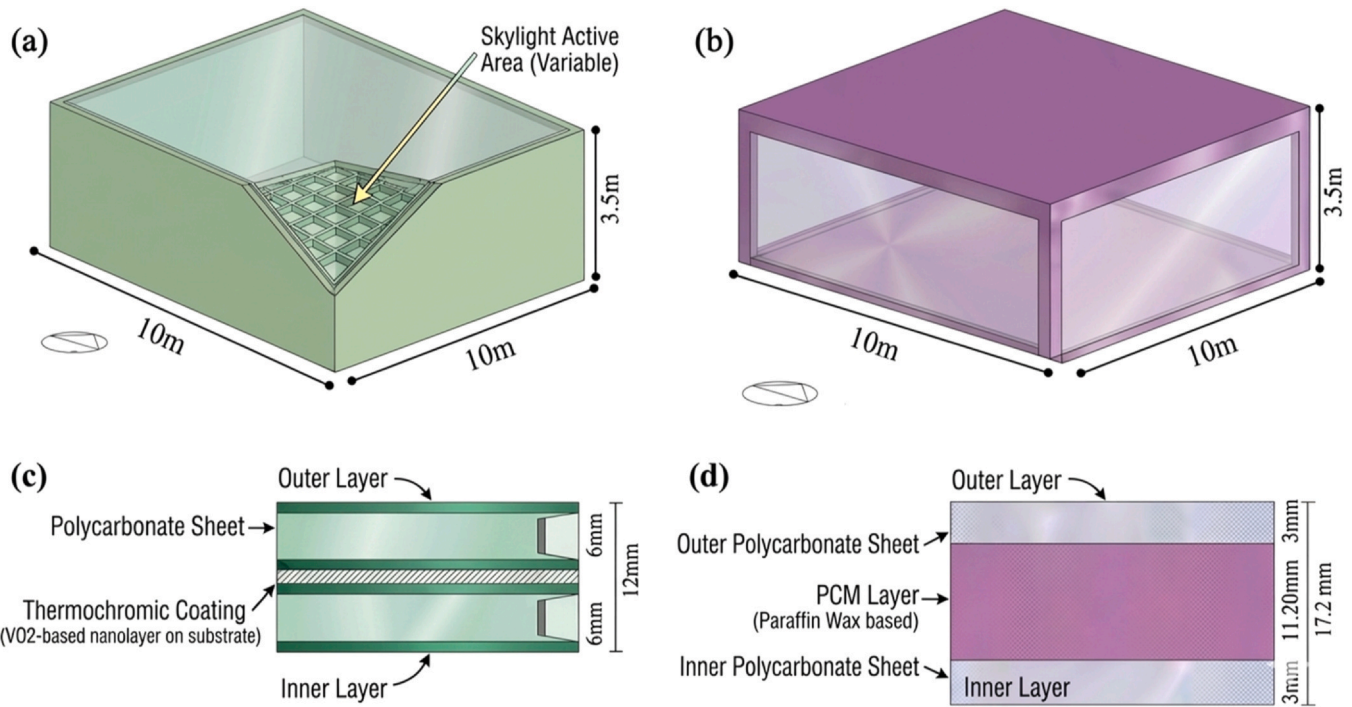


Fig. 5. Simulation models and advanced glazing cross-sections: (a) building model with variable skylight active area; (b) 100% WWR facade window configuration with opaque roof; (c) thermochromic assembly; and (d) PCM assembly.

and the centered RMSD divided by the observed standard deviation [70]. N is used as the denominator for both standard deviation and centered RMSD. The observed reference occupies $r = 1$, normalized standard deviation = 1, and normalized centered RMSD = 0. A shorter geometric distance from the reference indicates stronger combined agreement in correlation and variability. Centered RMSD removes the mean and evaluates mean bias and distributional agreement separately through descriptive statistics and the Kruskal-Wallis test. In Eq. (4), (O_i) and (P_i) denote actual and predicted cooling loads, (r) is the Pearson correlation coefficient, (σ_n) is the predicted-to-observed standard-deviation ratio, and (E'_n) is the normalized centered RMSD. The radial axis represents (σ_n) , while the angular coordinate is $\theta = \cos^{-1}(r)$. The observed reference is located at $(r = 1)$, $(\sigma_n = 1)$, and $(E'_n = 0)$. Lower (E'_n) values indicate closer agreement in variability and observation-level patterns. Because centering removes mean bias, the Kruskal-Wallis test evaluated distributional and mean agreement separately.

$$r = \frac{\sum_{i=1}^N (O_i - \bar{O})(P_i - \bar{P})}{\sqrt{\sum_{i=1}^N (O_i - \bar{O})^2} \sqrt{\sum_{i=1}^N (P_i - \bar{P})^2}},$$

$$[4pt] \sigma_n = \frac{\sigma_P}{\sigma_O}, \sigma_O = \sqrt{\frac{1}{N} \sum_{i=1}^N (O_i - \bar{O})^2}, \sigma_P = \sqrt{\frac{1}{N} \sum_{i=1}^N (P_i - \bar{P})^2},$$

$$[4pt] E'_n = \frac{1}{\sigma_O} \sqrt{\frac{1}{N} \sum_{i=1}^N [(P_i - \bar{P}) - (O_i - \bar{O})]^2} = \sqrt{1 + \sigma_n^2 - 2\sigma_n r},$$

$$[4pt] \theta = \cos^{-1}(r). \quad (4)$$

Monthly cooling energy defines the present ML target because the analysis evaluates seasonal EUI, annual performance, and PCM ranking. Monthly aggregation preserves these endpoints but removes melt-freeze cycling, incomplete solidification, heat-flux delay, cooling peaks, and TC threshold crossings. A standard 8760-h dataset would contain 78,840 rows per application for nine configurations, although adjacent rows remain autocorrelated and do not expand material diversity. The current ML dataset contains PCM cases only. Hourly TC prediction requires a separate dataset containing pane temperature, switching state, and dynamic transmittance or SHGC. A state-aware hourly model should include PCM enthalpy or liquid fraction, glazing-surface temperature, solar radiation, heat flux, lagged cooling load, and chronological validation. Uribe et al. [71] reported distinct reductions of 9.1% in annual energy and 10.5% in peak

cooling load from PCM glazing. Wang et al. [72] linked TC performance to transition time and hourly solar fluctuations. Hourly modeling therefore supports transient and peak-demand assessment, but increased row count does not guarantee higher accuracy. The present results apply to monthly cooling-energy prediction.

3.4.1. Model performance metrics

The evaluation framework applies seven complementary metrics to each set of paired actual and predicted cooling-load values (Table 5). For each application, the test set contains $N = 27$ pairs. In the following equations, O_i represents the actual cooling load, P_i represents the predicted cooling load, N represents the number of test observations, $\bar{O}$ and $\bar{P}$ represent the corresponding means, σ_O and σ_P represent their standard deviations, and r represents the Pearson correlation coefficient. RMSE and MAE retain the cooling-load unit of kWh. R^2 , NSE, CA, and KGE are dimensionless, while PBIAS reports percentage bias.

Çıtakoğlu et al. [76], Simsek et al. [77], and Bayram and Çıtakoğlu [78], the evaluation applies categorical performance bands to NSE, R^2 , KGE, and PBIAS. $NSE \leq 0.40$ indicates unsatisfactory performance, $0.40 < NSE \leq 0.50$ indicates acceptable performance, $0.50 < NSE \leq 0.65$ indicates satisfactory performance, $0.65 < NSE \leq 0.75$ indicates good performance, and $0.75 < NSE \leq 1.00$ represents the highest performance category. KGE follows corresponding limits of ≤ 0.40 , $0.40-0.50$, $0.50-0.60$, $0.60-0.70$, and $0.70-1.00$. For R^2 , values ≤ 0.40 indicate unsatisfactory performance, $0.40 < R^2 \leq 0.60$ indicates regular performance, $0.60 < R^2 \leq 0.80$ indicates good performance, and $0.80 < R^2 \leq 1.00$ represents the highest category. PBIAS = 0 indicates no aggregate bias. Absolute PBIAS values $\leq 10\%$, $10-15\%$, $15-25\%$, and $> 25\%$ indicate the highest, good, fair, and inadequate categories, respectively. The PBIAS sign identifies bias direction. Positive values indicate over-prediction, while negative values indicate underprediction. CA ranges from 0 to 1, with values approaching 1 indicating stronger agreement.

3.4.2. Hyperparameter configuration and model selection

Both application datasets contained 108 observations. The workflow created 81 training and 27 test observations through “test_size=0.25”, “shuffle=True”, and “random_state=42”. The “ColumnTransformer”

Table 5

Model performance metrics, description, and equations.

Performance Metrics	Equation(s)
Coefficient of determination, R^2 Description: R^2 compares squared prediction error with the total variation of the actual values. $R^2 = 1$ indicates exact prediction, $R^2 = 0$ represents the observed-mean benchmark, and $R^2 < 0$ indicates performance below this benchmark.	$1 - \frac{\sum_{i=1}^N (O_i - P_i)^2}{\sum_{i=1}^N (O_i - \bar{O})^2}$
Root mean square error, RMSE Description: RMSE expresses prediction error in kWh and gives larger residuals greater influence through squaring. Lower values indicate stronger accuracy, while RMSE = 0 denotes exact prediction.	$\sqrt{\frac{1}{N} \sum_{i=1}^N (O_i - P_i)^2}$
Mean absolute error, MAE Description: MAE reports the average absolute prediction error in kWh. The metric assigns linear weight to every residual, which supports direct interpretation of typical error magnitude. MAE = 0 denotes exact prediction.	$\frac{1}{N} \sum_{i=1}^N O_i - P_i $
Nash-Sutcliffe Efficiency, NSE Description: NSE compares squared prediction error against the variation around the actual mean [73]. NSE = 1 denotes exact prediction, NSE = 0 represents the observed-mean benchmark, and NSE < 0 indicates inferior performance relative to this benchmark.	$1 - \frac{\sum_{i=1}^N (O_i - P_i)^2}{\sum_{i=1}^N (O_i - \bar{O})^2}$
Coefficient of agreement, CA Description: CA follows Willmott's index of agreement for continuous predictions [74]. The metric compares squared residual error with potential error around the actual mean. CA ranges from 0 to 1. A value of 1 denotes exact agreement, while values approaching 0 indicate weak agreement.	$1 - \frac{\sum_{i=1}^N (P_i - O_i)^2}{\sum_{i=1}^N (P_i - \bar{O} + O_i - \bar{O})^2}$
Kling-Gupta Efficiency, KGE Description: KGE combines linear correlation, variability agreement, and mean agreement within one measure [75]. The correlation component r evaluates observation-level correspondence, α evaluates the predicted-to-actual variability ratio, and β evaluates the predicted-to-actual mean ratio. KGE = 1 denotes exact agreement. Departures in any component reduce the score.	$\frac{\bar{P}}{\bar{O}}$
Percent bias, PBIAS Description: PBIAS measures aggregate signed prediction bias relative to the total actual cooling load. PBIAS = 0 denotes no aggregate bias. Under the stated sign convention, positive values indicate underprediction and negative values indicate overprediction. Smaller absolute values indicate closer mean agreement.	$PBIAS(\%) = 100 \frac{\sum_{i=1}^N (P_i - O_i)}{\sum_{i=1}^N O_i}$

passed seven numerical variables without scaling and encoded “PCM_Family” through “OneHotEncoder” with drop = “first”. The Pipeline fitted the preprocessing stage within each training subset before model fitting. The workflow applied identical preprocessing and model settings to the skylight and window datasets. Table 6 reports the complete configurations used for test-set evaluation and five-fold cross-validation.

The analysis used a fixed-configuration selection strategy and no automated hyperparameter search. The workflow specified each setting before test-set evaluation and kept the settings constant across both applications. Random Forest used 300 trees, while Extra Trees used 500 trees to reduce finite-ensemble variation. A minimum leaf size of 2 prevented singleton terminal nodes in both ensemble models. Gradient Boosting combined 300 boosting stages, a learning rate of 0.05, and a maximum tree depth of 3 to control update size and individual-tree complexity. Distance-weighted k-Nearest Neighbors used five neighbors and assigned greater influence on nearer training observations. Five-fold KFold cross-validation used “shuffle = True” and “random_state = 42” to assess generalization across alternate data partitions. The held-out test set supplied the final comparison through R^2 , RMSE, MAE, NSE, CA, KGE, and PBIAS. Gradient Boosting recorded the strongest combined performance and therefore supported the subsequent SHAP analysis.

3.5. Software and computational environment

The building-energy workflow used EnergyPlus for simulation, while all post-simulation analysis used Python executed in Google Colab notebooks. Python handled data preprocessing, feature

engineering, model training, five-fold cross-validation, performance evaluation, visualization, and SHAP analysis. ChatGPT Codex (OpenAI) supported code drafting and structuring under researcher supervision. All scripts relied on study-specific simulated datasets and underwent manual verification before execution. Prior studies show that large language model-based coding assistants, including GitHub Copilot, support code generation and software development tasks under human oversight and improve development efficiency in structured workflows [79]. The workflow included five regression algorithms, 75/25 train-test splitting, five-fold cross-validation for the Skylight and Window datasets, 50-fold-level model runs, 10 final model evaluations, and two SHAP interpretability analyses. A local workstation supported data management and result processing. The system included an Intel Core i9-14900HX processor at 2.20 GHz, 32.0 GB RAM (31.7 GB usable), and an NVIDIA GeForce RTX 4070 Laptop GPU with 8 GB memory. Google Colab provides cloud-based execution resources, so local GPU specifications do not represent the runtime accelerator. No execution logs contained verified wall-clock times or persistent cloud hardware identifiers, so the study reports computational structure without runtime estimates.

3.6. Scope and limitations

The simulation results are subject to the assumptions of the EnergyPlus representation. Thermochromic switching and PCM phase change are described through parameterized thermo-optical and thermal properties. This formulation does not explicitly capture

Table 6

Hyperparameter configurations applied to the skylight and window datasets.

Model	Configuration
Linear Regression	fit_intercept = True, positive = False
Random Forest	n_estimators = 300, criterion = “squared_error,” max_depth = None, min_samples_split = 2, min_samples_leaf = 2, max_features = 1.0, bootstrap = True, random_state = 42
Gradient Boosting	loss = “squared_error,” n_estimators = 300, learning_rate = 0.05, subsample = 1.0, criterion = “friedman_mse,” min_samples_split = 2, min_samples_leaf = 1, max_depth = 3, max_features = None, random_state = 42
Extra Trees	n_estimators = 500, criterion = “squared_error,” max_depth = None, min_samples_split = 2, min_samples_leaf = 2, max_features = 1.0, bootstrap = False, random_state = 42, n_jobs = -1
k-Nearest Neighbors	n_neighbors = 5, weights = “distance,” algorithm = “auto,” leaf_size = 30, metric = “minkowski,” p = 2

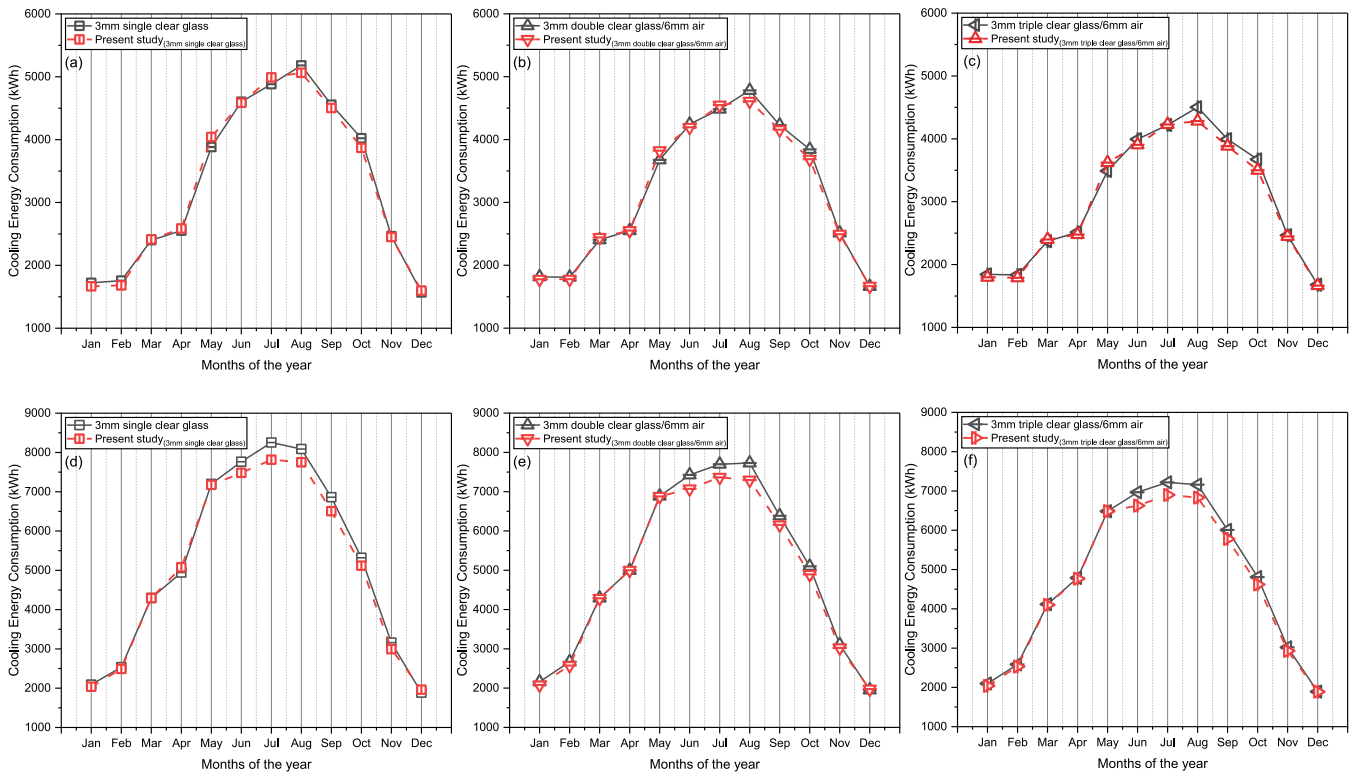


Fig. 6. Validation of energy consumption (cooling energy, kWh) for Windows (a-c), and Skylight (d-f).

manufacturing variability, glazing-edge effects, switching hysteresis, PCM leakage, optical aging, repeated-cycle degradation, or long-term material-property changes. Validation against standard simulation criteria supports model consistency but does not replace laboratory or field measurements.

Machine-learning methods also have distinct structural limitations. Linear Regression assumes additive linear relationships and cannot represent nonlinear interactions unless they are explicitly specified. Random Forest and Extra Trees can model nonlinear responses but generate piecewise predictions and generally extrapolate poorly beyond the training range. Gradient Boosting is sensitive to tree depth, learning rate, ensemble size, and sample size, whereas k-Nearest Neighbors depends on feature scaling, neighborhood density, and adequate local coverage. Although fixed configurations, a held-out test set, and five-fold cross-validation provide a consistent comparison, the relatively small simulation-derived datasets limit the certainty with which model rankings can be generalized.

The statistical and explainability methods require careful interpretation. The Kruskal-Wallis test evaluates differences in rank distributions and does not establish equality of means or physical superiority. Centered Taylor-diagram statistics emphasize correlation, standard deviation, and centered root-mean-square error but do not represent mean bias. SHAP values describe the contribution of predictors to the fitted model output and should therefore be interpreted as model associations rather than causal effects. Correlated climatic and envelope variables can also distribute attribution among related predictors, making individual SHAP rankings sensitive to the predictor set and sampling design.

4. Results and analysis

4.1. Validation of cooling energy consumption for skylight and window configurations

Fig. 6(a-c) compares the simulated and published monthly cooling-energy profiles for the single-, double-, and triple-clear window

systems. The simulations reproduced the seasonal increase toward the July-August peak and the subsequent decline. For the single-clear window, the simulated and published annual cooling energies were 1284.6 and 1289.8 kWh, respectively, corresponding to an NMBE of -0.40% and a CV(RMSE) of 2.72%. For the double-clear window, the corresponding values were 1176.3 and 1186.3 kWh, with an NMBE of -0.84% and a CV(RMSE) of 2.89%. For the triple-clear window, the simulated and published values were 1091.7 and 1112.0 kWh, respectively, producing an NMBE of -1.76% and a CV(RMSE) of 3.47%. These results indicate minor overall underprediction, with the maximum absolute monthly deviation remaining below 4.90%.

Fig. 6(d-f) presents the corresponding skylight comparisons. The simulated and published annual cooling energies were 1742.8 and 1691.4 kWh for the single-clear skylight, 1603.5 and 1551.0 kWh for the double-clear skylight, and 1487.2 and 1440.4 kWh for the triple-clear skylight. The resulting NMBE values were 3.04%, 3.38%, and 3.25%, while the CV(RMSE) values were 4.65%, 4.57%, and 4.28%, respectively. The higher simulated cooling energies reflect the greater solar exposure of horizontal skylights. The positive NMBE values indicate modest overall overprediction, and the maximum absolute monthly deviation remained below 5.97%. The close agreement between simulated and published cooling-energy values across all six configurations supports the use of the validated model for the subsequent thermochromic and PCM performance comparisons.

4.2. Performance of thermochromic material in skylight and vertical glazing systems

Fig. 7 shows a consistent reduction in cooling energy demand when thermochromic (TC) glazing is applied in both skylight and vertical configurations due to its temperature-dependent solar modulation. For skylights, TC glazing reduces cooling loads by approximately 31%, 27.7%, and 24.8% compared to single, double, and triple-glazing, respectively, with the largest reductions occurring during summer (May-August), where monthly cooling demand decreases to about 4500–5000

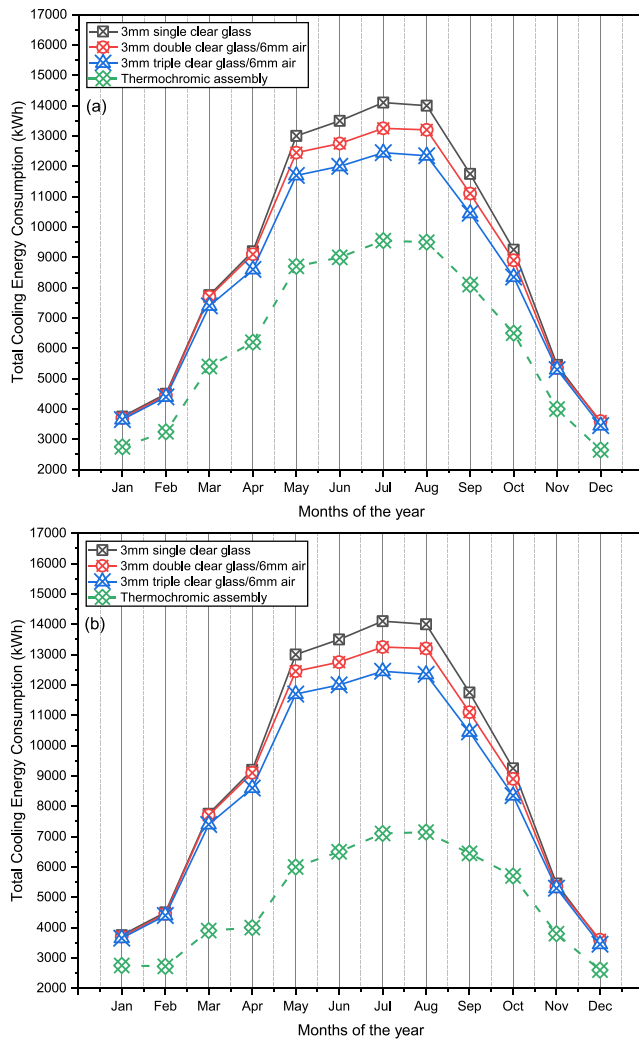


Fig. 7. Comparison of monthly cooling consumption for Conventional glazing vs. Thermochromic assembly, a) Skylight, and b) Window configurations.

kWh relative to single-clear glazing, confirming effective solar heat gain control under high irradiance.

TC glazing reduces vertical-window cooling energy from 998.89 to 1092.60 kWh/m²/year for clear glazing to 582.69 kWh/m²/year, producing savings of 41.7–46.7% and absolute reductions of 416.20–509.91 kWh/m²/year. For skylights, TC glazing produces savings of 24.8–31.0%. This percentage gap does not isolate orientation because each ratio uses a different reference glazing load. The difference also reflects the reference U-value and SHGC, glazing distribution, solar-incidence profile, and duration above the thermochromic transition temperature.

TC glazing reduces solar transmittance after thermal switching. Vertical façades receive direct radiation across multiple azimuths and daily periods, which extends the periods of effective solar modulation. Horizontal skylights receive intense near-normal midday radiation. Finite transmittance and inward heat release from solar absorption persist in the tinted state, leaving a larger residual cooling fraction. Mohammed et al. [67] reported a 60.1% higher annual cooling load for single-clear skylights than for wall glazing under the same Dhahran model. Yang et al. [80] reported orientation-specific reductions in summer solar-discomfort duration from 60.9% to 96.4% for thermochromic windows. These findings support an orientation-sensitive response, although hourly solar-gain and pane-temperature outputs remain necessary for direct causal decomposition.

4.3. PCM performance in skylight and vertical glazing systems

All PCM cooling load values reported in this section refer to the annual cooling energy demand attributed to the PCM-integrated glazing configuration under full-building simulation conditions. Fig. 8(a–d) shows that all nine PCM configurations exhibit similar seasonal trends, with July–August peaks, but their annual cooling loads differ significantly.

The PCM results are reported using two temporal aggregations. Fig. 8a & c illustrate each simulated month, whereas Fig. 8b & d reports the arithmetic mean across 12 months. Annual cooling energy is obtained by summing the monthly values. For the 100 m² skylight area, the nine PCM configurations produce mean monthly cooling-energy intensities of 21.96–36.72 kWh/m²/month, equivalent to annual cooling-energy intensities of 263.49–440.65 kWh/m²/year. InfiniteRPCM25C gives the lowest annual intensity, whereas WINCO Technologies ENERCEL 23 gives the highest. During May–August, individual PCM skylight loads range from 28.39 to 69.70 kWh/m²/month, and configuration-specific four-month means range from 37.76 to 63.14 kWh/m²/month. The TC range of approximately 45–50 kWh/m²/month therefore cannot be compared directly with the PCM annual values. The corrected comparison indicates configuration-dependent peak-season performance rather than uniformly lower PCM cooling demand.

Physically, these results do not demonstrate that PCM primarily reduces only the load remaining after solar rejection. PCM does not act as a sequential solar-rejection layer; it modifies the transient heat flux through latent-heat storage and release after the glazing absorbs and transmits part of the incident radiation. Consequently, it can reduce and delay conductive heat transfer to the zone, but it does not necessarily reduce incident solar gain or transmitted solar radiation itself. Because the present outputs are monthly cooling loads, they cannot partition the total response into transmitted solar gain, conductive heat flow, latent storage, and hourly peak demand. The defensible conclusion is therefore that PCM provides configuration-dependent thermal buffering and load attenuation, while its effects on peak solar gain and residual cooling load require hourly heat-balance or surface-flux analysis.

4.4. Comparative analysis of TC and PCM cooling energy consumption

Under the common 100 m² simulation basis, TC glazing reduced vertical-window annual cooling EUI to 582.69 kWh/m²/year, representing savings of 41.7–46.7% relative to the conventional glazing systems; the corresponding skylight savings were 24.8–31.0%. Annualizing the PCM data by summing the 12 monthly observations produced cooling EUIs of 263.49–440.65 kWh/m²/year for skylights and 432.03–474.83 kWh/m²/year for vertical windows. Infinite R PCM 25 °C achieved the lowest skylight EUI, whereas M51/Q25 achieved the lowest vertical-window EUI; WINCO ENERCEL 23 °C and M27/Q29 produced the respective maximum values. The findings therefore support an application-specific comparison rather than a universal claim that TC outperforms PCM: TC performance arises from temperature-triggered solar-transmittance modulation, whereas PCM performance reflects phase-change timing and the combined sensible and latent storage characteristics of each product. Because only melting temperatures of 23, 25, 27, and 29 °C were represented and the associated thermophysical properties were not varied independently, the results identify the best tested configurations but do not establish a universally optimal melting-temperature range.

4.5. Machine-learning regression framework for PCM

The 12 monthly outputs for each of the nine PCM configurations were reorganized into 108 PCM-month observations for each application, preserving the simulated monthly cooling-load targets without

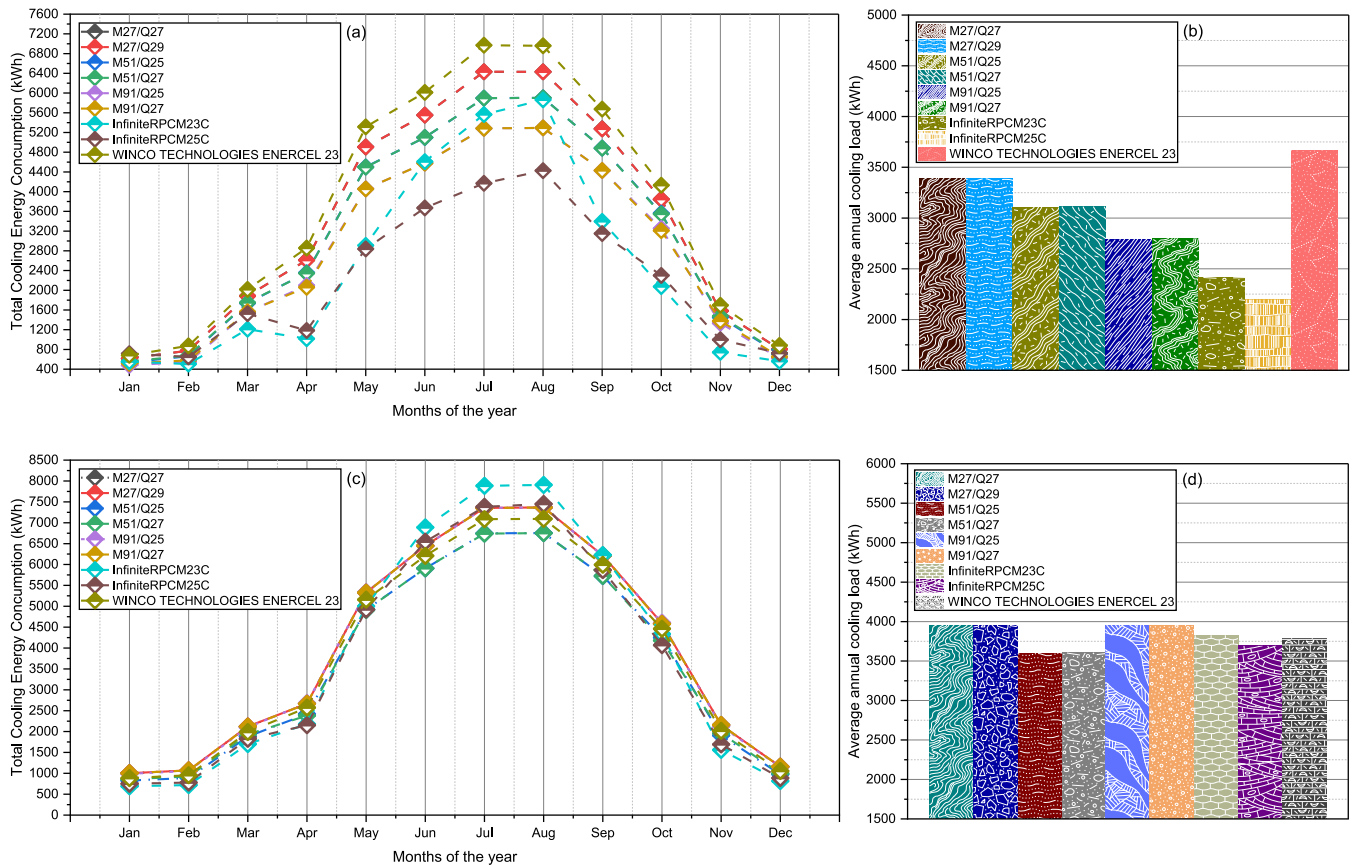


Fig. 8. Comparative evaluation of PCM configurations based on average annual and monthly cooling demands: (a-b) skylight application metrics and (c-d) vertical window application metrics.

creating additional independent simulation cases. Cooling loads ranged from 4.79 to 69.70 kWh/m²/month for skylights and from 6.90 to 79.05 kWh/m²/month for vertical windows. The predictors comprised mean outdoor temperature, relative humidity, daily sunshine duration, cooling degree days referenced to 24 °C (CDD-24), cyclical month encodings, PCM melting temperature, and material family; the three families were one-hot encoded with BioPCM as the reference category. Linear Regression, Random Forest, Gradient Boosting, Extra Trees, and k-Nearest Neighbors were evaluated through a common preprocessing pipeline. A reproducible 75/25 split using a random state of 42 allocated 81 observations to training and 27 to testing, with predictive performance assessed using R^2 , RMSE, and MAE. Shuffled five-fold cross-validation over the complete 108-observation dataset provided an additional measure of performance stability. Because records from the same months and PCM configurations could occur across different folds, the resulting metrics characterize interpolation within the evaluated Dhahran dataset rather than performance for previously unseen PCM assemblies or climatic conditions.

4.5.1. Test-set and cross-validation model performance

Fig. 9 positions the five regression models according to their test-set accuracy and error for vertical-window PCM cooling-load prediction, with the upper-left region representing the preferred combination of higher R^2 and lower RMSE. Gradient Boosting occupies this optimal zone, achieving $R^2 \approx 0.9897$, RMSE ≈ 252.8 kWh, and MAE ≈ 190 kWh, followed by Extra Trees ($R^2 = 0.988$; RMSE = 266.3 kWh) and Random Forest ($R^2 = 0.9876$; RMSE = 277.2 kWh). Although its R^2 advantage over Linear Regression ($R^2 = 0.982$; RMSE = 332.5 kWh) appears numerically small, Gradient Boosting reduces unexplained variance by approximately 42.5% and RMSE by about 24%, demonstrating a practically meaningful improvement.

The concentration of the ensemble models toward the upper-left region indicates that nonlinear relationships and interactions among climate, season, and PCM descriptors contribute to cooling demand; Gradient Boosting captures these effects effectively by sequentially correcting residual errors. Linear Regression still provides a strong benchmark, confirming that the dominant climate-load relationship is broadly systematic, but its higher residual error indicates unrepresented nonlinear structure. k-Nearest Neighbors records the weakest performance ($R^2 = 0.981$; RMSE = 340.4 kWh), suggesting that local similarity is less effective for the relatively small and sparsely distributed PCM-month dataset. Bubble sizes are scaled only for visualization, and the model ranking represents interpolation within the evaluated Dhahran cases rather than universal performance for unseen materials or climates.

Fig. 10 compares holdout and five-fold cross-validation performance for skylight Fig. 10(a-c) and window configuration Fig. 10(d-f) cooling-load prediction. Gradient Boosting leads the skylight models, with test/CV values of $R^2 = 0.970/0.974$, RMSE = 349.3/306.9 kWh, and MAE = 262.9/243.5 kWh, whereas kNN records the largest errors and lowest R^2 . Window predictions are consistently more accurate: Gradient Boosting achieves the best test-set performance ($R^2 = 0.990$, RMSE = 252.8 kWh, MAE = 189.8 kWh), while Extra Trees provides the highest CV R^2 (0.991) and lowest CV RMSE (220.5 kWh); Gradient Boosting retains the lowest CV MAE (172.3 kWh). The broad consistent test and cross-validation patterns support stable interpolation within the evaluated dataset, while the larger skylight errors indicate greater sensitivity to direct solar exposure and PCM-dependent load variation. These results do not establish performance for unseen PCM assemblies or climates because the folds contain related PCM-month observations.

Following the multi-criteria prediction assessment of Çitakoglu et al. [81], Table 7 reports seven complementary accuracy indicators for each

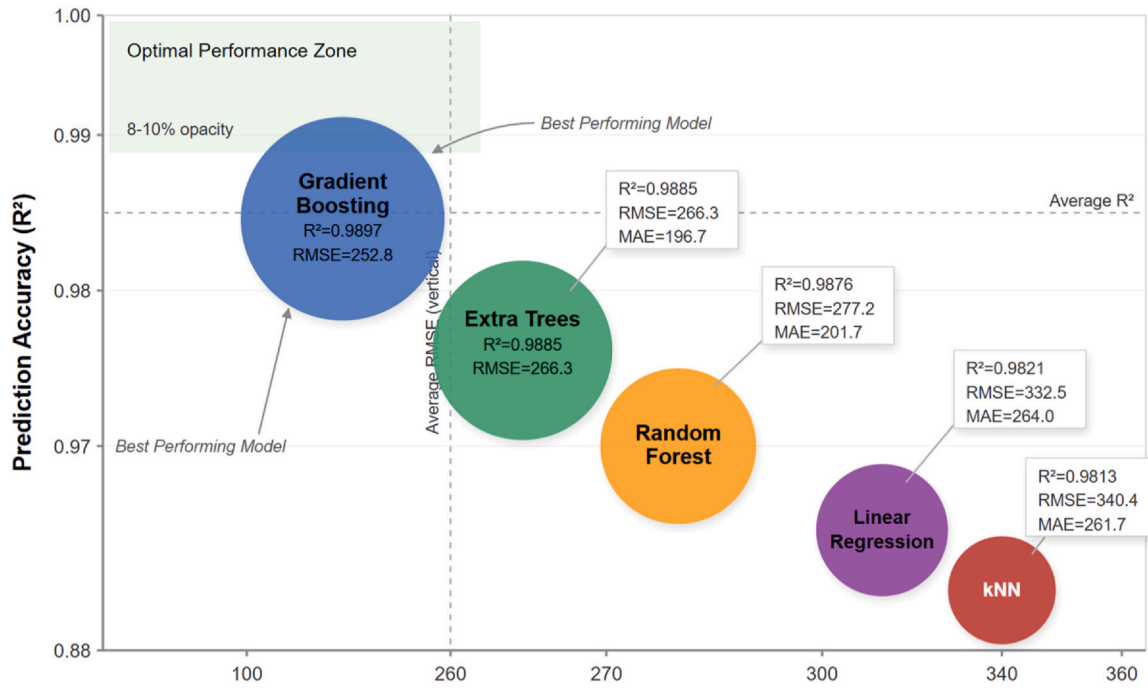


Fig. 9. Performance landscape of the evaluated machine-learning models for PCM cooling-load prediction.

held-out prediction set. Each calculation uses the same 27 paired actual and predicted observations. R^2 and NSE take equal values under the equations in Table 3, so the interpretation treats both metrics as one residual-efficiency result.

For skylights, Gradient Boosting records the lowest RMSE and MAE and the highest R^2 /NSE, CA, and KGE. For windows, Gradient Boosting records the lowest RMSE and MAE and the highest R^2 /NSE and CA. Random Forest records the highest window KGE, whereas Extra Trees

records the smallest absolute window PBIAS. The combined indicators identify Gradient Boosting as the most consistent model across both applications.

4.5.2. Kruskal-Wallis comparison of actual and predicted cooling loads

Fig. 11 compares the actual and predicted test-set cooling loads for both applications. For skylights, the actual mean reached 3007.16 kWh. Predicted means ranged from 2946.67 kWh for Linear Regression to

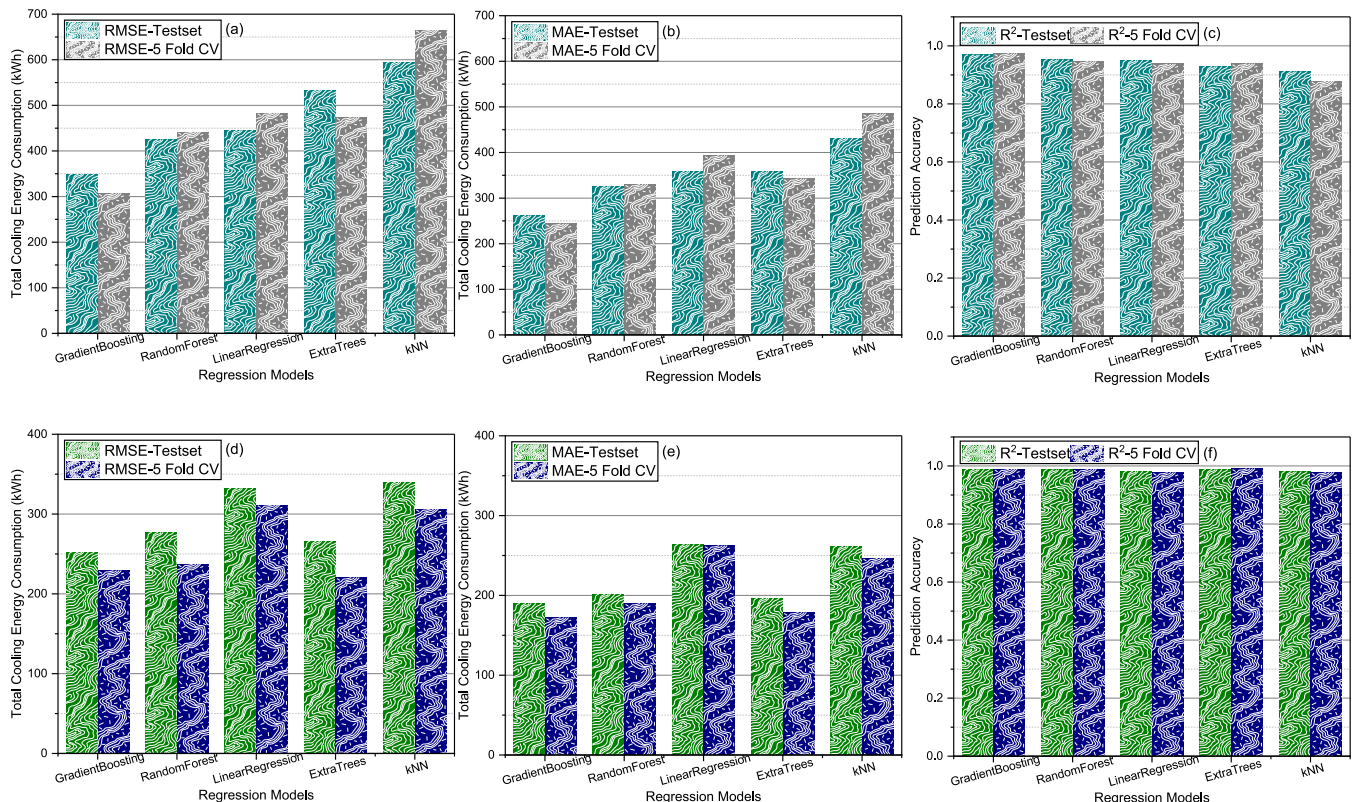


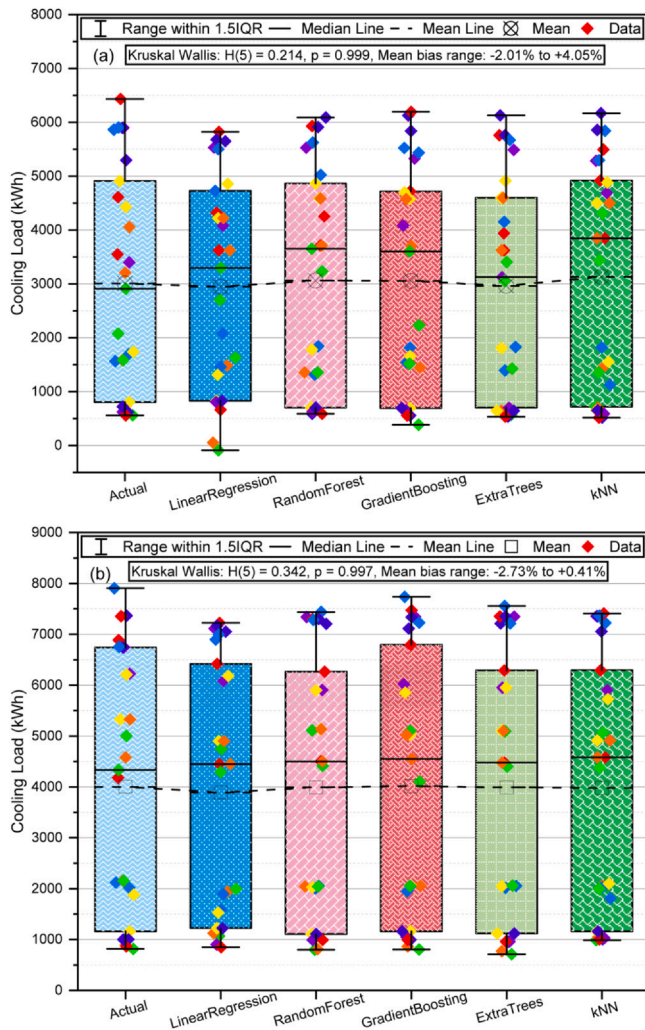
Fig. 10. Test-set vs. 5-fold cross-validation metrics across ML models: (a-c) skylight and (d-f) window cooling-load predictions evaluated via RMSE, MAE, and R^2 .

Table 7

Test-set accuracy metrics for skylight and window cooling-load predictions, N = 27 per application.

Application	Model	R ²	RMSE (kWh)	MAE (kWh)	NSE	CA	KGE	PBIAS (%)
Skylight	Linear Regression	0.9507	445.92	359.13	0.9507	0.9871	0.9532	-2.012
	Random Forest	0.9531	435.26	326.33	0.9531	0.9880	0.9671	+1.728
	Gradient Boosting	0.9698	349.31	262.94	0.9698	0.9924	0.9777	+1.625
	Extra Trees	0.9294	534.06	358.97	0.9294	0.9813	0.9445	-1.490
	k-Nearest Neighbors	0.9121	595.74	431.96	0.9121	0.9780	0.9403	+4.051
Window	Linear Regression	0.9821	332.52	264.01	0.9821	0.9954	0.9652	-2.734
	Random Forest	0.9876	277.24	201.72	0.9876	0.9969	0.9928	-0.268
	Gradient Boosting	0.9897	252.82	189.85	0.9897	0.9974	0.9893	+0.406
	Extra Trees	0.9885	266.33	196.74	0.9885	0.9971	0.9920	-0.264
	k-Nearest Neighbors	0.9813	340.40	261.74	0.9813	0.9952	0.9801	-0.472

Positive PBIAS indicates overprediction. Negative PBIAS indicates underprediction.

**Fig. 11.** Actual and predicted cooling-load distributions with means, medians, observations, and Kruskal-Wallis results, a) Skylights, b) Windows.

3128.97 kWh for k-Nearest Neighbors, corresponding to relative mean biases from -2.01% to $+4.05\%$. The Kruskal-Wallis test found no statistically detectable rank-distribution difference among the actual and predicted groups, $H(5) = 0.214$, $p = 0.999$. For windows, the actual mean reached 4000.15 kWh. Predicted means ranged from 3890.80 kWh for Linear Regression to 4016.38 kWh for Gradient Boosting, with relative mean biases from -2.73% to $+0.41\%$. The test also found no statistically detectable rank-distribution difference, $H(5) = 0.342$, $p = 0.997$. Both p values exceeded $\alpha = 0.05$. The descriptive means and relative biases show close mean agreement. The rank-based

tests found no statistically detectable separation among the actual and model-predicted distributions.

4.5.3. Taylor-diagram comparison of regression models

Fig. 12 presents the Taylor-diagram comparison for the test sets. For skylights, Gradient Boosting occupied the closest position to the observed reference, with $r = 0.9853$, normalized standard deviation = 1.0041, and normalized centered RMSD = 0.1721. Random Forest followed at 0.2151, Linear Regression at 0.2199, Extra Trees at 0.2649, and k-Nearest Neighbors at 0.2902. For windows, Gradient Boosting again produced the smallest normalized centered RMSD, 0.1014, with $r = 0.9949$ and normalized standard deviation = 1.0085. Extra Trees followed at 0.1070, Random Forest at 0.1113, Linear Regression at 0.1262, and k-Nearest Neighbors at 0.1366. All window models achieved $r \geq 0.9907$ and normalized standard deviations from 0.9800 to 1.0085. The lower normalized centered RMSD range for windows, 0.1014–0.1366, compared with 0.1721–0.2902 for skylights, indicates closer reproduction of observed variation across the window test set. Gradient Boosting achieved the strongest combined agreement for both applications.

4.5.4. Heatmap interpretation of test-set model accuracy

Fig. 13 consolidates the test-set metrics and identifies Gradient Boosting as the most accurate model for both applications, achieving RMSE = 349.31 kWh, MAE = 262.94 kWh, and $R^2 = 0.9698$ for skylights, and RMSE = 252.82 kWh, MAE = 189.85 kWh, and $R^2 = 0.9897$ for windows. Its window RMSE and MAE are approximately 28% lower than its skylight errors, confirming that vertical-window loads are more predictable. Random Forest and Extra Trees approach Gradient Boosting for windows, whereas Extra Trees deteriorates substantially for skylights (RMSE = 534.06 kWh; $R^2 = 0.9294$), indicating application-sensitive performance. kNN produces the weakest predictions in both datasets, while Linear Regression remains competitive but retains larger residual errors. The wider dispersion among skylight metrics reflects greater model sensitivity to its more variable PCM-dependent cooling response; therefore, Gradient Boosting provides the most consistent basis for subsequent interpretation within the evaluated cases.

4.6. SHAP-based model interpretability analysis

SHAP analysis was used to quantify the contribution of each predictor to the model output, as shown in Fig. 14. The results indicate that outdoor temperature and cooling degree days are the dominant predictors for both skylight and window applications. This ranking is physically consistent because these variables represent the primary seasonal thermal forcing and determine the absolute level of monthly cooling demand. The dominance of climate variables does not imply that PCM selection is unimportant. Rather, PCM family and melting temperature modify the climate-driven cooling-load baseline within the range of configurations evaluated.

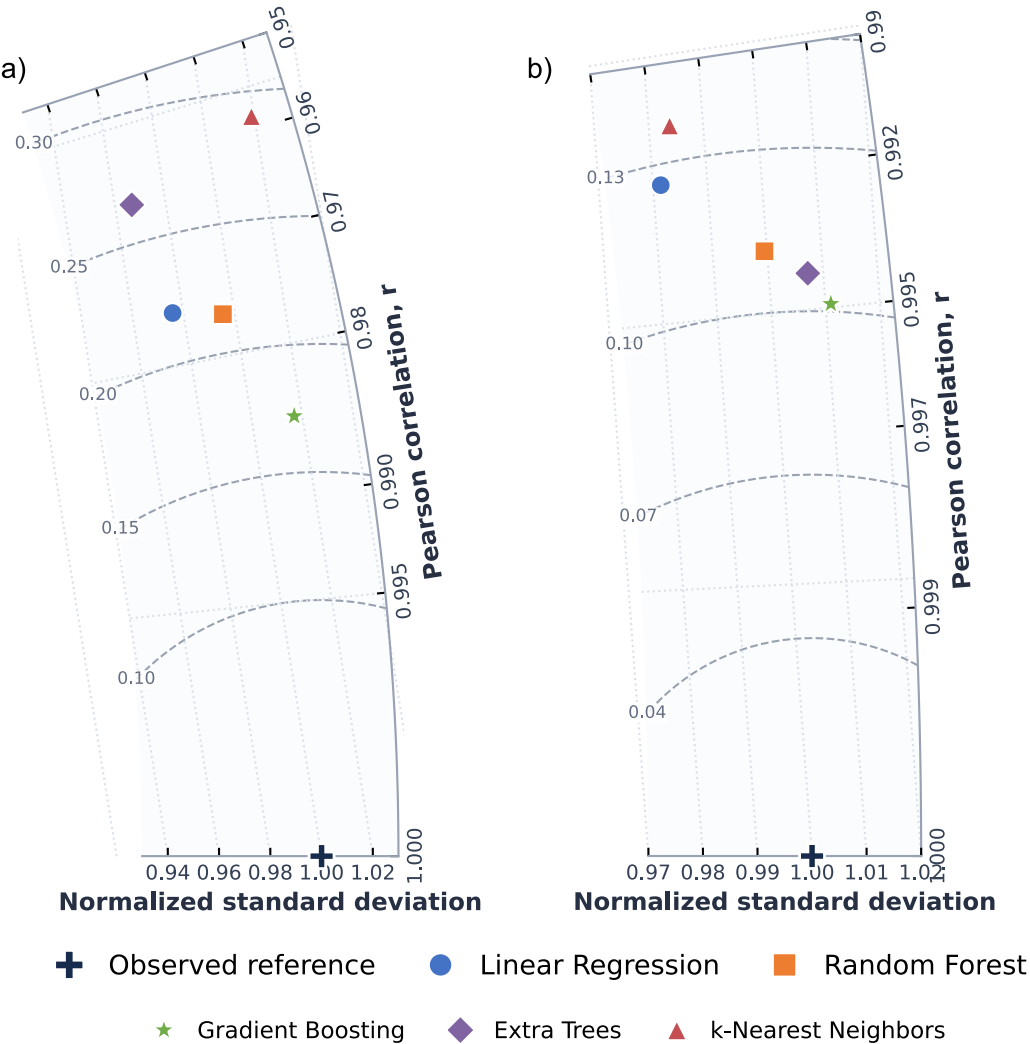


Fig. 12. Taylor diagram comparison of regression models for a) Skylight and b) Window configurations.

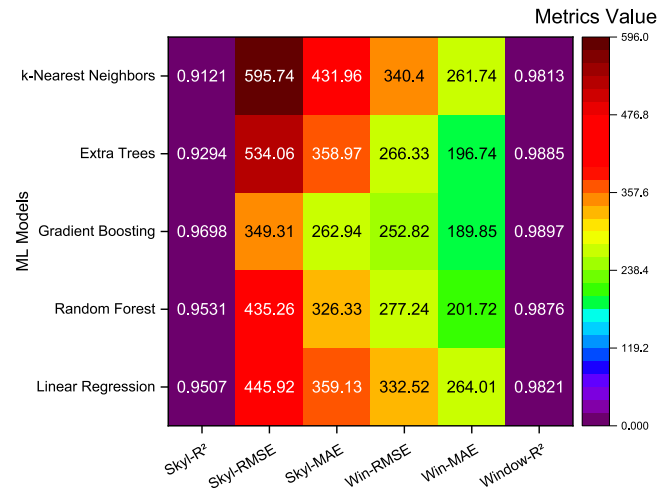


Fig. 13. Heatmap comparison of test-set RMSE, MAE (kWh), and R² for skylight and window PCM cooling-load prediction across ML models.

For the skylight application, the predicted cooling loads across the nine PCM configurations range from approximately 2200 to 3700 kWh, corresponding to a spread of about 1500 kWh. For the window application, the corresponding range is approximately 3600–4000 kWh, with a spread of about 400 kWh. Thus, PCM selection produces a substantially

larger variation in the skylight case than in the window case. The SHAP results are consistent with this pattern: PCM-family contributions reach approximately 500 kWh for skylights and approximately 300 kWh for windows. These values confirm that PCM variables have a meaningful application-dependent effect, even though their individual mean absolute SHAP values are lower than those of the climate predictors.

The relative SHAP ranking is also affected by the structure of the dataset. The analysis includes nine PCM configurations repeated over 12 months, whereas the climatic variables vary across the full annual cycle. As a result, the sampled variability of the climate predictors is broader than the variability represented by the selected PCM configurations. In addition, PCM behavior is represented through PCM family and melting temperature; thickness, latent-heat capacity, thermal conductivity, encapsulation, and other material properties are not included as independent predictors. Because PCM family and melting temperature are related descriptors of the tested configurations, their effects may also be distributed across the corresponding SHAP attributions.

Accordingly, the SHAP results should be interpreted as conditional importance within the simulated Dhahran building and the nine evaluated PCM configurations, rather than as a universal ranking of climate and material effects. The results show that climate drivers control the absolute cooling-load level, while PCM selection provides a meaningful secondary modification, particularly for skylights. Future work should expand the material-property space and apply grouped SHAP analysis to quantify the combined contribution of PCM-related variables more comprehensively.

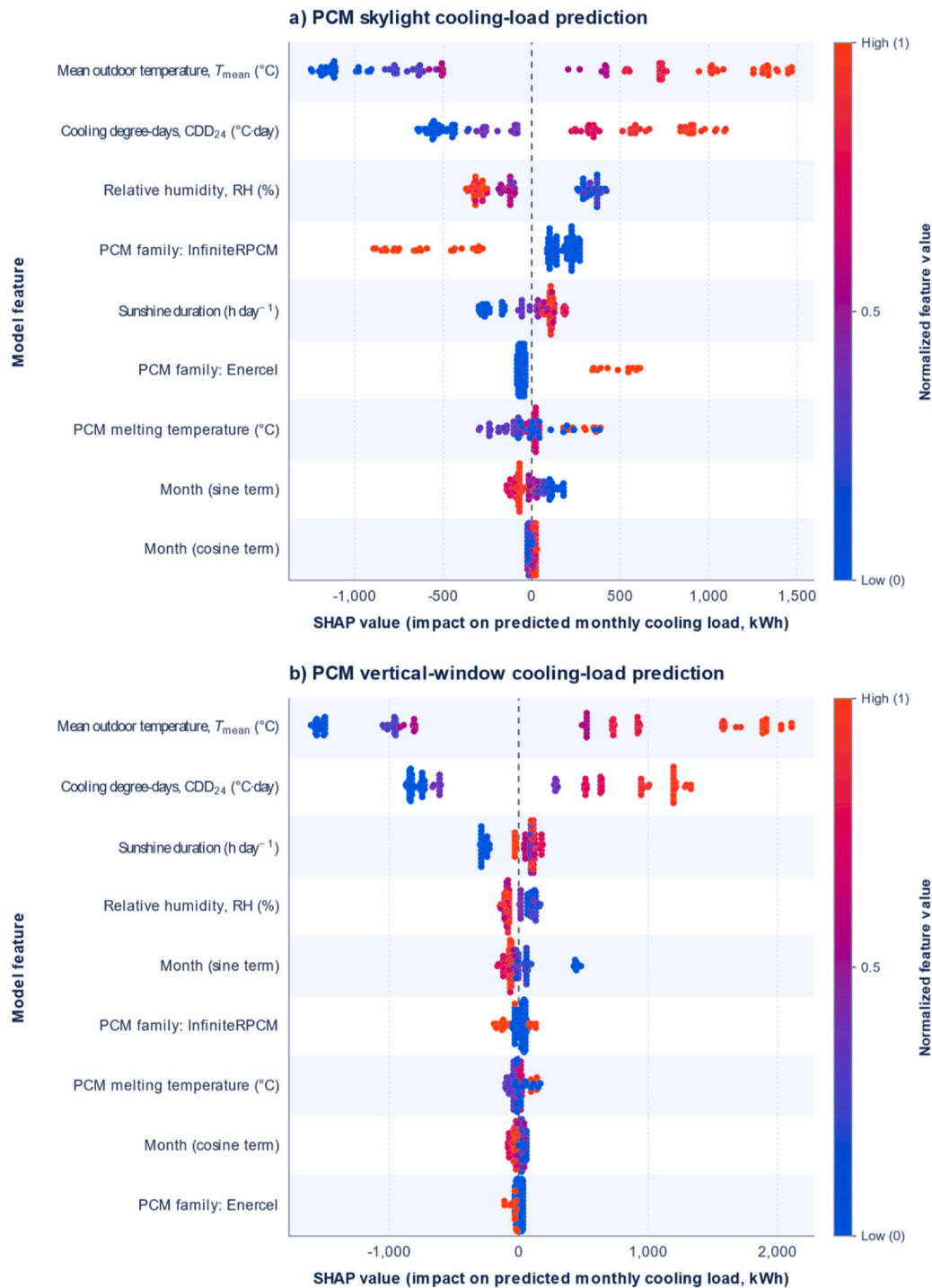


Fig. 14. SHAP feature contributions to PCM glazing cooling-load predictions.

5. Discussion

The validation results support comparison of the glazing alternatives under consistent boundary conditions. Conventional-window simulations showed NMBE of -1.76% to -0.40% and CV(RMSE) of $2.72\text{--}3.47\%$, whereas skylight simulations showed NMBE of $3.04\text{--}3.38\%$ and CV (RMSE) of $4.28\text{--}4.65\%$. These errors and the reproduced seasonal trends indicate that baseline-model uncertainty is unlikely to drive the differences among the glazing systems. However, the validation covers conventional glazing only and does not directly validate the TC or PCM assemblies.

TC glazing reduced cooling energy by $41.7\text{--}46.7\%$ for vertical windows and $24.8\text{--}31.0\%$ for skylights. The reductions reflect both

orientation and reference-glazing properties. When pane temperature reaches the thermochromic transition range, reduced solar transmittance limits inward solar gains. Vertical façades receive radiation across multiple orientations and periods, whereas skylights receive intense near-normal radiation around midday. The tinted skylight still transmits part of the incident radiation and transfers absorbed heat inward, which explains its higher residual cooling load.

PCM performance depends on transient latent-heat storage rather than steady conduction alone. Despite similar corrected U-values of $3.638\text{--}3.930\text{ W/m}^2\cdot\text{K}$, the skylight configurations produced annual cooling-energy intensities of $263.49\text{--}440.65\text{ kWh/m}^2/\text{year}$. PCM thermophysical properties and melting-range alignment explain this

variation. Repeated melting during high loads and solidification during cooler periods are necessary to reuse the latent-storage capacity. Monthly outputs, however, cannot resolve phase fraction, hourly peak shifting, or storage-release timing.

The machine-learning results indicate nonlinear interactions among climate, season, and PCM configuration. Gradient Boosting provided the strongest predictive agreement, with lower normalized centered RMSD for windows than skylights (0.1014 versus 0.1721). Greater skylight variability follows from stronger direct-solar exposure and higher sensitivity to PCM selection. The models nevertheless interpolate the 108 simulated cases and should not be extended to untested materials or boundary conditions.

Linear Regression provides a strong benchmark for window predictions, but the absolute R^2 difference understates the error reduction. Gradient Boosting raises R^2 from 0.9821 to 0.9897 and reduces unexplained variance by 42.5%. RMSE declines from 332.52 to 252.82 kWh, while MAE declines from 264.01 to 189.85 kWh, corresponding to reductions of 24.0 and 28.1%. The absolute PBIAS magnitude also decreases from 2.734% to 0.406%. Similar building-energy research reports higher cooling-load accuracy from Gradient Boosting than from Multiple Linear Regression [82]. Comparative evidence also shows that greater model complexity does not guarantee a meaningful accuracy advantage in every building-energy application [83]. These gains justify Gradient Boosting for final PCM comparison and SHAP interpretation within the sampled domain. Linear Regression remains suitable for transparent preliminary screening and coefficient-based explanation. Gradient Boosting trains offline using 81 observations and 300 depth-3 trees. Designers then query the fitted model during option assessment. Retained run records contain no method-specific wall-clock measurements, so the analysis makes no quantified runtime comparison. The practical benefit arises from lower error and nonlinear response representation rather than faster execution.

The SHAP analysis supports this interpretation but does not establish causality. Outdoor temperature and CDD24 dominate because they represent the annual cooling-driving conditions. PCM family and melting temperature contribute less strongly but remain influential, reaching approximately 500 kWh for skylights and 300 kWh for windows. Their larger skylight influence agrees with the wider configuration-dependent cooling-load range. Correlation between temperature and CDD24, together with related PCM thermophysical properties within the family variable, may distribute shared effects among features. The ranking therefore applies to the Dhahran climate and evaluated PCM set.

The findings have broader implications for façade selection and simulation-informed design. TC and PCM should not be treated as universally interchangeable smart-glazing solutions: TC directly modulates transmitted solar radiation and is therefore most advantageous when solar gain dominates, whereas PCM effectiveness depends on climate-specific alignment between its phase-change range and daily thermal-regeneration cycle. The simulation-ML-SHAP workflow can thus serve as a transparent screening tool that identifies influential variables and narrows candidate designs before detailed simulation or testing. Future research should experimentally validate the leading configurations; examine hourly heat fluxes across multiple climates, orientations, glazing ratios, and operating conditions; and vary PCM conductivity, latent heat, thickness, and enthalpy response independently. Integrating daylight, comfort, condensation, durability, life-cycle cost, and embodied carbon would ultimately support transferable multi-criteria selection rather than cooling-energy ranking alone.

6. Conclusions

This study developed a simulation-informed explainable machine-learning framework to compare thermochromic (TC) glazing with nine PCM-integrated assemblies in vertical-window and skylight applications under Dhahran conditions. The baseline EnergyPlus model reproduced published monthly conventional-glazing profiles with NMBE

values of -1.76% to -0.40% for windows and $3.04\text{--}3.38\%$ for skylights, supporting consistent comparative simulation. TC glazing produced the greatest cooling-energy reductions, achieving savings of $41.7\text{--}46.7\%$ for vertical windows and $24.8\text{--}31.0\%$ for skylights relative to the respective conventional systems. For the 100 m^2 skylight application, the PCM configurations yielded annual cooling-energy intensities of $263.49\text{--}440.65\text{ kWh/m}^2\text{/year}$. Infinite R PCM $25\text{ }^\circ\text{C}$ recorded the lowest value, confirming that PCM selection materially affects cooling performance.

Gradient Boosting reproduced the simulation outputs most accurately, achieving test-set R^2 values of 0.970 for skylights and 0.990 for windows, with RMSE values of 350.0 and 253.9 kWh, respectively. Its agreement across held-out, cross-validation, and Taylor-diagram evaluations supports its use as the final predictive model within the sampled domain. SHAP analysis identified mean outdoor temperature and CDD-24 as the principal predictors of monthly cooling demand, while PCM family and melting temperature provided smaller but meaningful application-dependent contributions. These attributions describe the fitted model and should not be interpreted as independent causal effects of individual material properties.

The principal contribution is an integrated workflow that combines validated building simulation, comparative smart-glazing assessment, machine-learning prediction, and model interpretation. It can support preliminary screening of glazing alternatives represented by the training data, but it does not establish a universal performance ranking. Future research should validate the leading TC and PCM configurations experimentally, resolve hourly switching and phase-change behavior, test additional climates and façade conditions, and vary PCM thermophysical properties independently. Including daylight, thermal comfort, durability, life-cycle cost, and embodied carbon would enable broader multi-criteria design decisions.

6.1. Study limitations and recommendations

The findings should be interpreted within the scope of the simulated Dhahran case study. The analysis considered one baseline building, one weather file, fixed operating conditions, nine PCM configurations, and monthly cooling-load outputs. As a result, the reported reductions do not establish universal performance rankings or confirm practical suitability under all building-use conditions. The study also did not evaluate heating demand, daylight and glare, thermal comfort, condensation risk, maintenance requirements, or life-cycle economic and environmental performance. These factors may influence the final selection of thermochromic glazing and PCM configurations.

The following recommendations are proposed:

- Validate the leading thermochromic and InfiniteRPCM25C configurations through controlled laboratory testing and long-term outdoor monitoring.
- Extend the analysis to multiple climates, building types, orientations, window-to-wall ratios, skylight ratios, shading conditions, internal-gain levels, and HVAC control strategies.
- Evaluate annual energy use and peak demand together with daylight, glare, thermal comfort, condensation, and operational performance.
- Compare alternatives using life-cycle cost, maintenance requirements, durability, and embodied-carbon indicators.
- Develop a decision framework that combines energy performance, occupant-related criteria, economic feasibility, and environmental impacts before practical implementation.

CRedit authorship contribution statement

Muhammad Asif: Writing – review & editing, Supervision, Resources. **Syed Yaqeen Mahdi:** Writing – review & editing, Validation, Software, Investigation. **Muhammad Huzaifa Zahid:** Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During manuscript development, AI tools assisted in refining the language, editing, and creating some graphical illustrations. Python-based machine-learning and visualization libraries generated scientific figures, including machine-learning performance visualizations and SHAP interpretability plots. This study encompasses all technical analyses, simulations, model development, data processing, figure verification, and scientific interpretations.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.nxener.2026.100997](https://doi.org/10.1016/j.nxener.2026.100997).

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